

An Optimal Quantum Query Lower Bound for Matrix-Product Verification

Shiva Kintali*

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Abstract

We prove that deciding whether $AB = C$ for three $n \times n$ matrices has bounded-error quantum query complexity $\Omega(n^{5/3})$ in the exact whole-entry oracle model, over every fixed nonzero associative ring with identity. Our lower bound matches the Buhrman–Špalek quantum-walk upper bound, establishing its optimality in the exact whole-entry query model. Our main idea is residual transfer: after averaging a bounded function of coordinate degree L over factors with fixed residual $AB - C$, we approximate its residual dependence by a sum of terms, each supported on $O(L/n)$ rows and columns. Equality-matrix substitution then transfers element-distinctness hardness.

1 Introduction, main results, and query model

Matrix-product verification asks whether $AB = C$ without computing the product. Freivalds’s randomized test uses $O(n^2)$ field operations [9]. We prove that the $O(n^{5/3})$ quantum query bound is optimal over every fixed nonzero associative ring with identity, combining a common source-hardness mechanism with distinct finite-field and integer masking constructions.

Related work. Buhrman and Špalek [6] improved an unpublished $O(n^{7/4})$ algorithm of Ambainis, Buhrman, Høyer, Karpinski, and Kurur to $O(n^{5/3})$ queries. Their stored-submatrix query algorithm works over rings. Their original time-efficient analysis is stated for integral domains [6]; extensions to rings are discussed in [12]. They proved an $\Omega(n^{3/2})$ lower bound over $\mathbb{F}_2$ by embedding an OR of parities; this exponent also extends to nonzero unital rings [12]. They asked whether it could be improved and whether a single wrong entry is harder to detect [6, Section 6.3]. Their algorithm builds on quantum walks [3, 17, 15].

Positive-weight adversaries encounter a certificate barrier: every such bound for a total function is at most $\sqrt{C_0(f)C_1(f)}$ [16, Theorem 3.2]. Here a NO certificate consists of one row of A , one column of B , and one entry of C , whereas all $3n^2$ entries form a YES certificate. Thus this barrier is $O(n^{3/2})$. Our proof uses real coordinate-degree lower bounds [4, 1, 2] and negative-weight adversaries [10, 14]. Bun and Thaler [7] construct explicit dual polynomials for collision and element distinctness; Belovs [5] converts dual witnesses directly into adversaries. We use a finite linear program for the exact dual correction and give its assignment-isometry specialization explicitly.

The integer argument also uses geometry of numbers: Ball’s cube-slicing theorem [20, 21], the kernel-covolume identity, and Minkowski’s second theorem [19, Section 2]. Rectangular-unimodular density results [18] suggest the surjectivity step, but a finite prime sieve and quantitative short-lift

*Email: shiva.kintali@gmail.com Homepage: shivakintali.github.io

estimate are needed here to obtain an explicit $O(n)$ -bit hard restriction. Boolean OR–AND matrix multiplication is a different problem [8, 12]; none of our ring reductions identifies it with ordinary multiplication.

1.1 Exact entry access

Throughout, $[n] = \{1, \dots, n\}$, all logarithms are natural unless marked $\log_2$, and products use the indicated order $(AB)_{ij} = \sum_k A_{ik}B_{kj}$. A ring is fixed, associative, and has identity $1_R \neq 0$, unless a different hypothesis is stated. Let $\text{MPV}_R(A, B, C) = \mathbf{1}[AB = C]$. For the addressed list z of the $3n^2$ entries, the exact-value oracle is

$$O_z|t, b, w\rangle = |t, b + z_t, w\rangle, \quad t \in [3n^2], \quad b \in R. \quad (1)$$

The response register is $\ell_2(R)$, with basis indexed by ring elements. Translation is unitary, and conjugating by $b \mapsto -b$ implements the inverse oracle. Three separate entry oracles and this addressed oracle are equivalent up to a constant factor.

Algorithms are finite query circuits with input-independent unitaries and a final measurement. Finite-outcome intermediate measurements and classically controlled stopping are also allowed, with a finite history tree and a worst-case query cap. Exact arithmetic and equality tests between queries have no query cost. The notation $Q_{1/3}$ denotes the least worst-case number of queries with success at least $2/3$ on every required input. For promise algorithms, truncate all histories at their promised worst-case cap; this makes acceptance defined and bounded on all oracle inputs without changing promised behavior. On a finite restriction, the usual XOR oracle for fixed-length entry labels simulates (1) with two queries: write the label into a clean ancilla, translate the original response register by the decoded value, and erase the label. The same construction implements an inverse or controlled query and allows an arbitrary original response register. For infinite rings this is an ideal exact-value model: $\ell_2(R)$ may be nonseparable, and no finite-qubit, finite-time, or finite-precision implementation is asserted. The algorithms never require a uniform superposition over R .

All lower bounds have finite support. On a finite product restriction $\Omega = \prod_{t=1}^{3n^2} \Sigma_t$, let $V_L(\Omega)$ be the real span of functions depending on at most L entry coordinates. This is equivalently degree at most L in the one-hot indicators $\mathbf{1}[z_t = a]$. The bounded approximate coordinate degree $\widetilde{\deg}_{\text{bd}, 1/3}(f)$ is the least such degree of a real function p satisfying $0 \leq p \leq 1$ on the entire product domain and $|p - f| \leq 1/3$ on every required input. Function sup norms are taken over that entire domain, including off-promise inputs. This degree is not algebraic degree over the underlying ring. The finite negative-weight adversary value is defined in Section 5.4; for infinite R , $\text{Adv}^\pm(\text{MPV}_R)$ denotes its supremum over finite restrictions.

1.2 Main results

Theorem 1.1 (Optimal query complexity over unital rings). *For every fixed nonzero associative ring R with identity, in the exact whole-entry model,*

$$\text{Adv}^\pm(\text{MPV}_R) = \Omega(n^{5/3}), \quad Q_{1/3}(\text{MPV}_R) = \Theta(n^{5/3}).$$

The lower bound is witnessed on a finite restriction. Its implicit constants may depend on R .

The two arithmetic ingredients have stronger quantitative statements. For a finite field put $\Omega_{n,q} = (\mathbb{F}_q^{n \times n})^3$. For integers define

$$I_H = [-2H, 2H] \cap \mathbb{Z}, \quad J_{n,H} = [-4nH^2 - 1, 4nH^2 + 1] \cap \mathbb{Z},$$

$$\mathcal{X}_{n,H} = I_H^{n \times n} \times I_H^{n \times n} \times J_{n,H}^{n \times n}, \quad H_n = 2^{100n}.$$

In both settings write $f(A, B, C) = \mathbf{1}[AB = C]$.

Theorem 1.2 (Finite fields, uniformly in field size). *There are absolute $c > 0$ and n_0 such that for every finite field $\mathbb{F}_q$ and $n \geq n_0$,*

$$\widetilde{\deg}_{\text{bd},1/3}(f|_{\Omega_{n,q}}) \geq cn^{5/3}, \quad Q_{1/3}(f|_{\Omega_{n,q}}) = \Theta(n^{5/3}).$$

All constants are independent of q , which may grow with n .

Theorem 1.3 (Integers with linear entry bit length). *There are absolute $c > 0$ and n_0 such that for every $n \geq n_0$ and integer $H \geq H_n$,*

$$\widetilde{\deg}_{\text{bd},1/3}(f|_{\mathcal{X}_{n,H}}) \geq cn^{5/3}, \quad Q_{1/3}(f|_{\mathcal{X}_{n,H}}) = \Theta(n^{5/3}).$$

Taking $H = H_n$ gives hard integer inputs with $O(n)$ bits per entry.

Let E_{ij} have entry 1 at (i, j) and zero elsewhere, interpreted in the stated arithmetic domain.

Theorem 1.4 (Adversary, singleton, and distributional bounds). *In both settings of Theorems 1.2 and 1.3, the negative-weight adversary bound is $\Omega(n^{5/3})$, also under the promise*

$$AB - C \in \{0\} \cup \{E_{ij} : i, j \in [n]\}.$$

The same query and adversary lower bounds hold for the larger promise that at most one entry is wrong.

For the finite-field case, choose A, B independently and uniformly. For the integer case, choose all their entries independently and uniformly from $[-H, H] \cap \mathbb{Z}$. With probability $1/2$ set $C = AB$; otherwise choose (i, j) uniformly and set $C = AB - E_{ij}$. Average success at least $2/3$ under this balanced distribution requires $\Omega(n^{5/3})$ queries for algorithms with a worst-case query cap. Constants are uniform in q and in $H \geq H_n$, respectively.

These results include every field and $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$, with exact entry access as defined above. The bit-length claim concerns the hard restriction, not a bit-time upper bound. For singleton residuals in composite characteristic, see Section 6.

Contributions and proof strategy. The common mechanism averages $p(A, B, AB - D)$ while retaining the residual D . It maps bounded coordinate degree L to dependence on at most $K = O(L/n)$ rows and columns per residual Fourier term, with exponentially small uniform error. Substituting $D_{ij} = \mathbf{1}[x_i = y_j]$ gives source degree at most $2K$. A balanced split of an element-distinctness instance forces $K = \Omega(n^{2/3})$, hence $L = \Omega(n^{5/3})$. This proves the general-input lower bound without an adversary construction.

The two ways to obtain residual transfer are genuinely different. Over finite fields, character orthogonality and random systematic-code distance give exact one-sided support restrictions. Over the integers, a prime sieve and short lattice lifts make low-coordinate observations nearly invariant under residual shifts. A signed correction applied to each row before tensoring turns this into an exact support restriction without paying a whole-input decomposition norm. In both cases, two separate factor averages are combined by a controlled row projection; the factors are never jointly conditioned.

For the stronger conclusions, a collision sample produces a uniform zero-versus-singleton gap test. It controls one explicit signed seed; a finite LP corrects its low-degree moments, and the

assignment isometries of [5] specify a negative-weight adversary. The construction is finite but not claimed to be computationally efficient. These averaging steps act on bounded functions and are not query-preserving algorithmic reductions. Elementary entry-local encodings subsequently give the ring and structural consequences.

The transfer framework also yields $\Theta(ks^{2/3})$ queries for s -by- k times k -by- s verification when $k \geq s$, matching the rectangular upper bound of [6, Section 6.1]. Appendix A collects entry-local consequences for algebraic identities, Gram comparisons, and structured inputs. Their promises are separate and retain the exact-entry model.

Organization. Section 2 gives the common framework; Sections 3 and 4 supply its arithmetic ingredients. Section 5 proves the stronger bounds, Section 6 extends to rings, and Section 7 treats rectangular verification and localization. Appendix A contains the application constructions and detailed localization analysis.

2 The common residual-transfer framework

The framework in this section has no field-specific hypothesis. Its input is a uniform approximation of residual averaging by functions with restricted row and column support. Sections 3 and 4 verify that input by different arguments.

2.1 Finite alphabets and polynomial degree

Fix a finite product domain $\Omega = \prod_{t=1}^{3n^2} \Sigma_t$ and a predicate $f : \Omega \rightarrow \{0, 1\}$. A partial assignment α fixes values of some input coordinates; $|\alpha|$ is their number and $z \sim \alpha$ means agreement. The empty assignment is allowed. Abbreviate $V_L(\Omega)$ to V_L when the domain is specified.

Lemma 2.1 (Coordinate-degree filtration). *The space V_L is the real span of indicators $\mathbf{1}[z \sim \alpha]$ with $|\alpha| \leq L$, and is exactly the space of one-hot polynomial functions of degree at most L on valid alphabet inputs. Fixing coordinates cannot increase this degree.*

Proof. Any function depending on coordinates U has the assignment expansion

$$F(z) = \sum_{a \in \prod_{t \in U} \Sigma_t} F(a) \prod_{t \in U} \mathbf{1}[z_t = a_t].$$

Conversely, products of one-hot indicators vanish when inconsistent and otherwise reduce to an assignment indicator. Fixing coordinates turns their indicator factors into constants. $\square$

Lemma 2.2 (Acceptance degree on finite restrictions). *A T -query algorithm in the model of Section 1.1 has a globally $[0, 1]$ -valued acceptance function in V_{2T} on every finite product restriction, also with infinite-dimensional workspace.*

Proof. On the restriction, the oracle and its inverse are finite sums

$$O_z^{\pm 1} = \sum_t \sum_{a \in \Sigma_t} \mathbf{1}[z_t = a] |t\rangle\langle t| \otimes \tau_{\pm a},$$

with identities on the remaining registers, where τ_a is translation by a . Induction on queries expresses the final state as a finite sum of indicator monomials of degree at most T times fixed Hilbert-space vectors. For the fixed acceptance effect M , the probability $\langle \psi_z, M \psi_z \rangle$ is a real polynomial of degree

at most $2T$. Its range is $[0, 1]$ on all valid oracles. This is the finite-alphabet polynomial method [4], [2, Lemma 2.2]; no scalar coordinate basis for the workspace or infinite-domain averaging is needed.

For a finite adaptive history tree, apply the same induction to the unnormalized state along each history, including its fixed measurement operators. By retaining ancillary registers, initial states and measurement instruments may first be purified, so that each branch is represented by one fixed linear operator. A history with $t \leq T$ queries has state degree at most t and acceptance contribution of degree at most $2t$. Summing the finitely many accepting histories gives the same bound $2T$. No translation-invariant dummy response state or physical implementation of a controlled oracle is assumed. $\square$

We use the collision and element-distinctness degree bounds of Aaronson and Shi [1] in the polynomial formulation of [2, Definition 2.1 and Theorem 2.3]. Fix $0 < \varepsilon < 1/2$ and let N be even. If $M \geq 3N/2$, every real polynomial in the one-hot indicators $\mathbf{1}[u(i) = j]$ that lies in $[0, 1]$ on all maps $u : [N] \rightarrow [M]$, is at least $1 - \varepsilon$ on injective maps and at most ε on exactly two-to-one maps has degree $\Omega(N^{1/3})$. Every polynomial with the same global range bound that is at least $1 - \varepsilon$ on injective maps and at most ε on all non-injective maps has degree $\Omega(N^{2/3})$ when $M \geq N$; [2, Theorem 2.3] states this for $M = \Omega(N^2)$, and [2, Theorem 3.1] transfers the degree to every $M \geq N$. We use the element-distinctness bound with $M = 4N^2$. The implied constants depend only on ε . Crucially, boundedness is required on all legal maps, including off-promise maps, but not on invalid one-hot encodings.

2.2 Residual support and the transfer hypothesis

Let $\mathcal{D}_n = \{0, 1\}^{n \times n}$ and define $\mathcal{W}_K$ as the real span of functions depending only on $D[R, C]$, where $R, C \subseteq [n]$ and $|R|, |C| \leq K$. Equivalently, $\mathcal{W}_K$ is spanned by the Boolean characters

$$\chi_S(D) = (-1)^{\sum_{(i,j) \in S} D_{ij}}$$

whose supports meet at most K rows and at most K columns. Indeed every function on a rectangle has a Boolean Fourier expansion supported in that rectangle, and every such character depends on its row-by-column rectangle. In particular, constants belong to $\mathcal{W}_K$. This support description, rather than a choice of field character, is the common interface.

For each $D \in \mathcal{D}_n$, let μ_D be a probability measure on Ω supported on inputs satisfying $f(z) = \mathbf{1}[D = 0]$, and put

$$(\mathcal{R}p)(D) = \mathbb{E}_{z \sim \mu_D} p(z).$$

Thus $\mathcal{R}$ preserves constants and maps $[0, 1]$ -valued functions to $[0, 1]$ -valued functions. In the applications, μ_D is the law of $(A, B, AB - D)$ for the stated factor distribution. A Boolean residual is interpreted through the distinct ring elements $0, 1$; integer multiplication is unchanged.

Definition 2.3 (Transfer hypothesis). For every fixed $a > 0$ and all sufficiently large n , suppose that, simultaneously for $0 \leq L \leq an^{5/3}$, there is a real linear map

$$\mathcal{H}_L : V_L \longrightarrow \mathcal{W}_{\lfloor 16L/n \rfloor}$$

such that

$$\|\mathcal{R}p - \mathcal{H}_L p\|_\infty \leq \xi_{n,L} \|p\|_\infty, \quad 0 \leq \xi_{n,L} \leq e^{-cn} \quad (2)$$

for a constant $c > 0$. The domain norm is on all of Ω and the output norm on all of $\mathcal{D}_n$. When the construction has auxiliary parameters, the constants and thresholds must be uniform in those parameters for uniform conclusions.

Corollary 2.4 (Normalization). *Suppose Definition 2.3 holds. For $p \in V_L$ with $0 \leq p \leq 1$, write $g = \mathcal{R}p$, $h_0 = \mathcal{H}_L p$, $\xi = \xi_{n,L}$, $K = \lfloor 16L/n \rfloor$, and set*

$$h = \frac{h_0 + \xi}{1 + 2\xi}. \quad (3)$$

Then $h \in \mathcal{W}_K$, $0 \leq h \leq 1$, and $\|h - g\|_\infty \leq 2\xi/(1 + 2\xi)$. For any probability measure ν on residuals,

$$h(0) - \mathbb{E}_\nu h(D) \geq \frac{g(0) - \mathbb{E}_\nu g(D) - 2\xi}{1 + 2\xi}. \quad (4)$$

Proof. All pointwise inequalities here and below are on the indicated finite domain. Since $g \in [0, 1]$ and $\|h_0 - g\|_\infty \leq \xi$, the range of h_0 is contained in $[-\xi, 1 + \xi]$. The constant function belongs to $\mathcal{W}_K$, so the normalization preserves the support restriction. Moreover,

$$h - g = \frac{h_0 - g + \xi(1 - 2g)}{1 + 2\xi},$$

which proves the uniform estimate. In a difference of two expectations the added constant cancels, and the two errors are each at most ξ ; this proves (4). $\square$

2.3 Equality substitution and element distinctness

For source lists $x, y \in [M]^n$, put

$$D(x, y)_{ij} = \mathbf{1}[x_i = y_j] \in \{0, 1\}.$$

Source labels do not need to embed in the target ring.

Lemma 2.5 (Degree of the pullback). *For $h \in \mathcal{W}_K$, the function $h(D(x, y))$ has one-hot source-coordinate degree at most $2K$. If $0 \leq h \leq 1$ on $\mathcal{D}_n$, the same bounds hold for all list pairs, including lists with internal repetitions.*

Proof. Each function of $D[R, C]$ becomes a function only of the labels x_i , $i \in R$, and y_j , $j \in C$. Their number is at most $2K$, so Lemma 2.1 gives the degree bound term by term. Every list pair produces a Boolean residual, which proves boundedness. $\square$

Lemma 2.6 (Source hardness). *If $h \in \mathcal{W}_K$, $0 \leq h \leq 1$, $h(0) \geq 0.98$, and $h(D) \leq 0.02$ for every nonzero $D \in \mathcal{D}_n$, then $K = \Omega(n^{2/3})$ with absolute constants.*

Proof. By Lemma 2.5, on source lists $x, y \in [M]^n$, $h(D(x, y))$ is a degree-at-most- $2K$ approximant to the predicate that the lists share no value. Take $M = 16n^2 = 4(2n)^2$. For $z \in [M]^{2n}$ and a permutation σ of $[2n]$, put $x^\sigma = (z_{\sigma(1)}, \dots, z_{\sigma(n)})$ and $y^\sigma = (z_{\sigma(n+1)}, \dots, z_{\sigma(2n)})$, and define

$$F(z) = \mathbb{E}_\sigma h(D(x^\sigma, y^\sigma)),$$

the average over a uniformly random permutation σ . Each indicator $\mathbf{1}[x_i^\sigma = a]$ is the single one-hot indicator $\mathbf{1}[z_{\sigma(i)} = a]$ of z , and the positions $\sigma(1), \dots, \sigma(2n)$ are distinct, so for each fixed σ the summand has degree at most $2K$ in the one-hot indicators of z . The law of σ does not depend on z , so F has degree at most $2K$ and takes values in $[0, 1]$. All-distinct inputs give $F \geq 0.98$. If the input has a repeated value, fix one equal pair of distinct positions. The two positions land in different lists with probability

$$2 \frac{n}{2n} \frac{n}{2n-1} = \frac{n}{2n-1} > 1/2.$$

On that event a cross collision exists, so the value is at most 0.02; on the complement it is at most one. Therefore $F \leq 1 - 0.98n/(2n - 1) < 0.51$ on every colliding input. Its square satisfies

$$F^2 \geq 0.98^2 > 2/3 \quad \text{on distinct inputs}, \quad F^2 \leq 0.51^2 < 1/3 \quad \text{on colliding inputs},$$

and has degree at most $4K$. Since $0 \leq F \leq 1$ on every legal source map, the same is true of F^2 . Thus F^2 approximates element distinctness on $[M]^{2n}$ with error below $1/3$ in the sense of [2, Definition 2.1], is bounded on all inputs, and $M = 4(2n)^2$. The element-distinctness degree bound recalled above, applied with source length $N = 2n$, gives $4K = \Omega((2n)^{2/3}) = \Omega(n^{2/3})$. $\square$

Proposition 2.7 (Transfer to the general-input lower bound). *Under Definition 2.3,*

$$\widetilde{\deg}_{\text{bd}, 1/3}(f) = \Omega(n^{5/3}), \quad Q_{1/3}(f) = \Omega(n^{5/3}).$$

The constants are uniform whenever the transfer constants are uniform.

Proof. Choose a bounded approximant $p \in V_d$. For an absolute odd integer s with $e^{-s/18} \leq 0.01$, the majority polynomial

$$\text{Maj}_s(t) = \sum_{j=(s+1)/2}^s \binom{s}{j} t^j (1-t)^{s-j}$$

lies in $[0, 1]$ for $t \in [0, 1]$ and reduces error $1/3$ to 0.01 by Hoeffding's inequality [11]. Thus $p' = \text{Maj}_s(p) \in V_{sd}$ has these bounds globally. If $sd > n^{5/3}$ the degree conclusion already holds. Otherwise use the transfer with $L = sd$ and $a = 1$. For large n , its error is at most $1/200$. The support condition on μ_D gives $(\mathcal{R}p')(0) \geq 0.99$ and $(\mathcal{R}p')(D) \leq 0.01$ for $D \neq 0$. Corollary 2.4 therefore supplies h satisfying Lemma 2.6, with $K = \lfloor 16sd/n \rfloor$. Consequently $16sd/n \geq K = \Omega(n^{2/3})$, so $d = \Omega(n^{5/3})$. Lemma 2.2 gives the query lower bound. $\square$

3 Finite-field residual transfer

Fix a finite field $\mathbb{F}_q$. All constants in this section are absolute and independent of q . We first work on the entire field-valued residual space, then restrict to the Boolean cube to invoke Section 2.

3.1 Characters and coordinate degree

Choose a nontrivial additive character $\chi : \mathbb{F}_q \rightarrow \mathbb{C}$, for example $\chi(a) = \exp(2\pi i \text{Tr}_{\mathbb{F}_q/\mathbb{F}_\ell}(a)/\ell)$ when $q = \ell^s$ with ℓ prime (identifying the trace value with its standard integer representative in the exponential). The trace map is nonzero, hence the displayed character is nontrivial. For $u \neq 0$, multiplication by u permutes $\mathbb{F}_q$; character orthogonality therefore gives $\mathbb{E}_{a \in \mathbb{F}_q} \chi(ua) = \mathbf{1}[u = 0]$. For matrices put $\langle S, D \rangle = \sum_{i,j} S_{ij} D_{ij}$ and let $\text{wt}(S)$ count nonzero entries. The characters on $\Omega_{n,q}$ are

$$(A, B, C) \mapsto \chi(\langle U, A \rangle + \langle V, B \rangle + \langle S, C \rangle).$$

Write $V_L = V_L(\Omega_{n,q})$ for the coordinate filtration already defined in Section 2.

Lemma 3.1 (Fourier description of coordinate degree). *V_L consists precisely of the real-valued functions whose Fourier coefficients vanish whenever $\text{wt}(U) + \text{wt}(V) + \text{wt}(S) > L$.*

Proof. The one-coordinate identities

$$\mathbf{1}[z_t = a] = q^{-1} \sum_{u \in \mathbb{F}_q} \chi(u(z_t - a)), \quad \chi(uz_t) = \sum_{a \in \mathbb{F}_q} \chi(ua) \mathbf{1}[z_t = a]$$

show equality of the complex character and assignment spans at every support size. Taking real parts and applying Lemma 2.1 gives the stated description. $\square$

This degree is not finite-field algebraic degree. All polynomial lower bounds in this manuscript concern real functions and one-hot coordinates. No boundedness on invalid one-hot encodings is assumed or needed.

3.2 Random systematic codes

Set $\delta = 1/16$. Call B row-good if

$$\text{wt}(v) + \text{wt}(vB^\top) \geq \delta n \quad \text{for every nonzero row vector } v \in \mathbb{F}_q^n. \quad (5)$$

Call A column-good if

$$\text{wt}(v) + \text{wt}(A^\top v) \geq \delta n \quad \text{for every nonzero column vector } v \in \mathbb{F}_q^n. \quad (6)$$

These are distance conditions on systematic linear codes of length $2n$.

Lemma 3.2. *Put $\mathcal{H}(t) = -t \log t - (1-t) \log(1-t)$ and*

$$c_* = \frac{15}{16} \log 2 - 2\mathcal{H}(1/32) > 0, \quad \eta_n^F = e^{-c_* n}.$$

A uniform B fails (5) with probability at most η_n^F ; a uniform A fails (6) with the same bound.

Proof. For fixed nonzero v , the vector $vB^\top$ is uniform in $\mathbb{F}_q^n$, since its coordinates are independent nonzero linear functionals of the independent uniform rows of B . For each fixed $v \neq 0$ and each w , the probability of $vB^\top = w$ is therefore q^{-n} . Take a union bound over such pairs with $v \neq 0$ and concatenated weight at most $n/16$, and then enlarge their number by allowing $v = 0$. This gives

$$\begin{aligned} \Pr[B \text{ is not row-good}] &\leq q^{-n} \sum_{j=0}^{\lfloor n/16 \rfloor} \binom{2n}{j} (q-1)^j \\ &\leq \exp \left\{ -n \left(\log q - 2\mathcal{H}(1/32) - \frac{1}{16} \log(q-1) \right) \right\}. \end{aligned} \quad (7)$$

There are $\binom{2n}{j} (q-1)^j$ concatenated pairs of weight j . The first bound deliberately includes the non-failing boundary and the pairs with $v = 0$, so it remains an upper bound. The second uses the binomial entropy bound $\sum_{j \leq tm'} \binom{m'}{j} \leq e^{m' \mathcal{H}(t)}$ with $m' = 2n$ and $t = 1/32$, and $(q-1)^j \leq (q-1)^{n/16}$. Since $q \geq 2$ and $\log(q-1) \leq \log q$, $\log q - \frac{1}{16} \log(q-1) \geq \frac{15}{16} \log 2$, so this is at most $e^{-c_* n}$. Finally, $\mathcal{H}(t) \leq t \log(1/t) + t$ gives $c_* \geq (10 \log 2 - 1)/16 > 0$. The column case uses independent columns of A ; neither argument requires invertibility. $\square$

3.3 Transfer on field-valued residuals

For any real function p on $\Omega_{n,q}$, define

$$(\mathcal{R}p)(D) = \mathbb{E}_{A,B} p(A, B, AB - D), \quad D \in \mathbb{F}_q^{n \times n}, \quad (8)$$

where both factors are independent and uniform. For a residual frequency S , write $r(S)$ and $c(S)$ for its numbers of nonzero rows and columns. A function on residuals has *row Fourier degree* at most K if every frequency S in its Fourier support satisfies $r(S) \leq K$, and *column Fourier degree* at most K if every such S satisfies $c(S) \leq K$. Let $\mathcal{W}_K^{(q)}$ be the real vector space of real functions on residuals with both row and column Fourier degree at most K .

Theorem 3.3 (Uniform residual transfer). *For each integer $L \geq 0$, put*

$$K = \lfloor 16L/n \rfloor, \quad M_{n,K} = \sum_{j=0}^{\min(K,n)} \binom{n}{j} 2^j, \quad \zeta_{n,L} = \eta_n^F (1 + 2M_{n,K}).$$

There is a real linear map $\mathcal{H}_L : V_L \rightarrow \mathcal{W}_K^{(q)}$ such that

$$\|\mathcal{R}p - \mathcal{H}_L p\|_\infty \leq \zeta_{n,L} \|p\|_\infty \quad (p \in V_L). \quad (9)$$

*Moreover, for every $a > 0$ there is $n_1(a)$ such that $\zeta_{n,L} \leq e^{-c*n/2}$ for all $n \geq n_1(a)$ and all integers $0 \leq L \leq an^{5/3}$.*

Proof. Fix B and expand $p(A, B, \cdot)$ in the characters of (A, C) :

$$\begin{aligned} p(A, B, C) &= \sum_{U,S} a_{U,S}(B) \chi(\langle U, A \rangle + \langle S, C \rangle), \\ a_{U,S}(B) &= 0 \quad \text{if } \text{wt}(U) + \text{wt}(S) > L. \end{aligned} \quad (10)$$

Here the coefficients are normalized with respect to uniform measure:

$$a_{U,S}(B) = \mathbb{E}_{A,C} p(A, B, C) \chi(-\langle U, A \rangle - \langle S, C \rangle) = \sum_V \hat{p}(U, V, S) \chi(\langle V, B \rangle),$$

where $\hat{p}(U, V, S)$ denotes the Fourier coefficient of p on $\Omega_{n,q}$ at the frequency (U, V, S) . Since $p \in V_L$, every term of the last sum with $\text{wt}(U) + \text{wt}(S) > L$ vanishes, which justifies the vanishing condition in (10). The matrix identity $\langle S, AB \rangle = \langle SB^\top, A \rangle$ and orthogonality in the *unconditioned* uniform variable A give

$$g_B(D) := \mathbb{E}_A p(A, B, AB - D) = \sum_S a_{-SB^\top, S}(B) \chi(-\langle S, D \rangle). \quad (11)$$

By the vanishing condition in (10), a surviving residual frequency (one with a nonzero coefficient in (11)) satisfies

$$\text{wt}(S) + \text{wt}(SB^\top) \leq L. \quad (12)$$

If B is row-good, sum (5) over the nonzero rows of S . The resulting left side is exactly $\text{wt}(S) + \text{wt}(SB^\top)$, so it is at least $\delta nr(S)$. Since $r(S)$ is integral, (12) gives $r(S) \leq \lfloor 16L/n \rfloor = K$. The subnormalized average

$$g_{\text{row}}(D) = \mathbb{E}_B [\mathbf{1}[B \text{ row-good}] g_B(D)]$$

has row Fourier degree at most K . With $g = \mathcal{R}p$,

$$\|g - g_{\text{row}}\|_\infty \leq \eta_n^F \|p\|_\infty. \quad (13)$$

Indeed, the omitted set of B has probability at most η_n^F , and $|g_B(D)| \leq \|p\|_\infty$ uniformly in B, D .

Independently fix A and write

$$p(A, B, C) = \sum_{V,S} b_{V,S}(A) \chi(\langle V, B \rangle + \langle S, C \rangle), \quad b_{V,S}(A) = 0 \text{ if } \text{wt}(V) + \text{wt}(S) > L,$$

where the vanishing holds for the same reason as in (10). Since $\langle S, AB \rangle = \langle A^\top S, B \rangle$, the corresponding average is

$$\begin{aligned} g_{\text{col}}(D) &= \mathbb{E}_A [\mathbf{1}[A \text{ column-good}] \mathbb{E}_B p(A, B, AB - D)] \\ &= \mathbb{E}_A \left[\mathbf{1}[A \text{ column-good}] \sum_S b_{-A^\top S, S}(A) \chi(-\langle S, D \rangle) \right], \end{aligned}$$

where the inner B average is uniform and unconditioned. Every surviving frequency for a column-good A satisfies

$$\text{wt}(S) + \text{wt}(A^\top S) \leq L.$$

Summing (6) over the nonzero columns of S gives $\text{wt}(S) + \text{wt}(A^\top S) \geq \delta n c(S)$ and hence $c(S) \leq K$. Thus g_{col} has column Fourier degree at most K and

$$\|g - g_{\text{col}}\|_\infty \leq \eta_n^F \|p\|_\infty. \quad (14)$$

These are two separate, subnormalized averages. In each calculation the variable used for character orthogonality remains unconditioned; we have not conditioned both factors to be good.

For each row i of D , let E_i be uniform averaging over that entire row while holding the other entries fixed. These are commuting projections of L_∞ norm one. For $R \subseteq [n]$, define

$$Q_R = \prod_{i \in R} (I - E_i) \prod_{i \notin R} E_i, \quad P_K = \sum_{|R| \leq K} Q_R.$$

On a character $\chi(\langle S, D \rangle)$, E_i is the identity if row i of S is zero and is zero otherwise. Hence Q_R selects precisely row support R , and P_K selects row degree at most K . Also

$$\|P_K\|_{\infty \rightarrow \infty} \leq \sum_{|R| \leq K} 2^{|R|} = M_{n,K}. \quad (15)$$

This projection is not assumed to be an L_∞ contraction; its full $M_{n,K}$ cost is retained. No factor involving the row alphabet size q^n occurs. The projection is diagonal in the entrywise Fourier basis and only deletes frequencies. It therefore cannot introduce a frequency with more than K nonzero columns. Consequently $\mathcal{H}_{Lp} := P_K g_{\text{col}}$ has both required support restrictions. Since $P_K g_{\text{row}} = g_{\text{row}}$, the triangle inequality and (13)–(14) give

$$\begin{aligned} \|P_K g_{\text{col}} - g\|_\infty &\leq \|g_{\text{row}} - g\|_\infty + \|P_K(g_{\text{col}} - g_{\text{row}})\|_\infty \\ &\leq \eta_n^F \|p\|_\infty + M_{n,K}(2\eta_n^F \|p\|_\infty), \end{aligned}$$

which proves (9). The averages and row projections preserve real functions, proving the asserted real linearity.

For $L \leq an^{5/3}$, $K \leq 16an^{2/3} < n$ for all sufficiently large n . If $K \geq 1$, the crude estimate $M_{n,K} \leq (K+1)(2n)^K$ gives $\log M_{n,K} = O_a(n^{2/3} \log n) = o(n)$; if $K = 0$ the assertion is immediate. Consequently $\log(1 + 2M_{n,K}) \leq c_* n/2$ for all sufficiently large n , uniformly in this range of L . Multiplying by $e^{-c_* n}$ proves the stated error bound. $\square$

3.4 Completing the finite-field lower bound

Proof of Theorem 1.2. Choose A, B independently and uniformly over $\mathbb{F}_q^{n \times n}$, and restrict residual averaging and the operator of Theorem 3.3 to $\mathcal{D}_n$. A field character with at most K nonzero rows and columns depends only on the corresponding rectangle of residual entries. Its restriction to $\{0, 1\}^{n \times n}$ therefore lies in the complex span of the Boolean space $\mathcal{W}_K$. For a real-valued combination, taking real parts gives membership in the real space. No identification of the field character with a Boolean character is required; in particular, no condition on $\chi(1)$ is used.

Restriction cannot increase the supremum norm. Hence the two restricted operators satisfy Definition 2.3, uniformly in q . Every sampled triple has $AB - C = D$, so the required fiber condition also holds. Proposition 2.7 proves the coordinate-degree and query lower bounds. The matching upper bound is the stored-submatrix walk of [6]; its ring-independent query analysis is recalled in Section 6. $\square$

4 Integer lattice masking and residual transfer

We now use ordinary integer arithmetic and the outer product domain $\mathcal{X}_{n,H}$ of Section 1.2. Random factors are sampled from the inner box $[-H, H]^{n \times n} \cap \mathbb{Z}^{n \times n}$. This distinction is essential: the function norm and boundedness are on the outer domain, which also contains the translated arguments. The required width is $H \geq H_n = 2^{100n}$.

Unlike finite-field character averaging, integer averaging is only approximately invariant under a residual shift. We first construct short integer lifts uniformly over all sparse observations, then correct each row operator before tensoring. The arithmetic exceptional set and the much smaller translation error are controlled separately.

4.1 Effective surjectivity and short integer lifts

For a rank- d lattice Λ in its Euclidean span, $\det \Lambda$ is its covolume and $\lambda_i(\Lambda)$ is the least radius of a centered Euclidean ball containing i linearly independent lattice vectors. Let B_2^d be the Euclidean unit ball. We use Minkowski's second theorem in its standard form

$$\prod_{i=1}^d \lambda_i(\Lambda) \leq \frac{2^d}{\text{vol}(B_2^d)} \det \Lambda \quad (16)$$

for a rank- d Euclidean lattice; see the general lattice statement in [19, Section 2, equation (12)]. An asymptotic version of the arithmetic step below also follows from the rectangular-unimodular density theorem [18, Corollary 17]. We instead give an elementary finite-box estimate, needed to control the entry bit length.

Lemma 4.1 (Effective surjectivity). *Let T be a $k \times m$ matrix with independent uniform integer entries in $[-H, H]$, where $k \geq 1$, $m > 2k$, and $H \geq \max\{k, 4m\}$. Put $d = m - k$ and $W = 2H + 1$. Then*

$$\begin{aligned} \Pr[T\mathbb{Z}^m \neq \mathbb{Z}^k] &\leq 3 \cdot 2^{-d} + \mathcal{E}_{k,m}(H), \\ \mathcal{E}_{k,m}(H) &= \frac{k}{W} + \frac{2^{m+1}(4m)^d}{dH^d} + 2k \cdot 2^d W^{-(m-2k+1)}. \end{aligned} \quad (17)$$

Proof. For a prime p , a rank-deficient matrix over $\mathbb{F}_p$ has all its columns in one of at most $2p^{k-1}$ hyperplanes. For a fixed hyperplane, condition on all but one coordinate with nonzero normal coefficient. An interval of W integers puts mass at most $p^{-1} + W^{-1}$ on a specified residue. Thus

$$\Pr[\text{rank}(T \bmod p) < k] \leq 2p^{k-1}(p^{-1} + W^{-1})^m. \quad (18)$$

If $p \leq H/(2m)$, the right side is at most $2e^{1/4}p^{-d-1}$. Since $d \geq 2$,

$$\sum_{r=2}^{\infty} r^{-d-1} \leq 2^{-d}(1/2 + 1/d) \leq 2^{-d}.$$

The union over these small primes costs less than $3 \cdot 2^{-d}$.

For $H/(2m) < p \leq W$, (18) is at most $2^{m+1}p^{-d-1}$. With $R = \lfloor H/(2m) \rfloor \geq H/(4m)$, their total contribution is at most

$$2^{m+1} \sum_{r>R} r^{-d-1} \leq \frac{2^{m+1}}{dR^d} \leq \frac{2^{m+1}(4m)^d}{dH^d}.$$

For $p > W$, first expose the square matrix S formed by the first k columns. A nonzero polynomial of total degree k vanishes on at most a k/W fraction of a Cartesian grid with W choices per coordinate. Applying this grid root bound to the determinant gives $\Pr[\det S = 0] \leq k/W$. Condition on a nonzero determinant D . By Hadamard's inequality and $H \geq k$,

$$|D| \leq (\sqrt{k}H)^k < W^{2k}.$$

Thus fewer than $2k$ primes greater than W divide D . Any large prime causing rank failure must be among this list, which depends only on S .

Fix such a prime and expose the remaining d columns. While their span together with S is proper, the probability that a new column fails to increase its rank is at most $1/W$: one nonzero linear equation allows at most one of the $W < p$ possible pivot values. Define a failed step to be false once full rank has been reached. This bound then holds conditionally on every previous history. If final rank is still below k , at least $d - k + 1 = m - 2k + 1$ steps failed. Repeated conditional expectation and a union bound over their positions give probability at most $2^d W^{-(m-2k+1)}$. Union over the fewer than $2k$ candidate primes gives the final term of $\mathcal{E}_{k,m}(H)$.

Outside $\det S = 0$, the image of T has finite index in $\mathbb{Z}^k$. It is a proper subgroup exactly when it fails to have full rank modulo some prime, by Smith normal form. This proves (17). $\square$

We will also use the elementary kernel-covolume identity: for every surjective integer map $T : \mathbb{Z}^m \rightarrow \mathbb{Z}^k$,

$$\det(\ker T \cap \mathbb{Z}^m) = \sqrt{\det(TT^\top)}. \quad (19)$$

Indeed, choose an integer right inverse U and an integer basis matrix K of the kernel. The columns of $[K \ U]$ form a basis of $\mathbb{Z}^m$. In orthonormal coordinates splitting the real kernel and its orthogonal complement, the two diagonal-block determinants have absolute values $\det(\ker T \cap \mathbb{Z}^m)$ and $1/\sqrt{\det(TT^\top)}$. Their product is one.

Lemma 4.2 (Euclidean primitive-vector estimate). *Let T and W be as in Lemma 4.1, and put $d = m - k$. For a primitive nonzero $v \in \mathbb{Z}^m$, write $r_v = \|v\|_2$. If x is uniform in $[-H, H]^m \cap \mathbb{Z}^m$, then*

$$\Pr[x \cdot v = 0] \leq \frac{\sqrt{2}}{Wr_v} \exp\left(\frac{(m-1)^2 r_v}{W}\right). \quad (20)$$

Consequently, for every real $R \geq \sqrt{m}$,

$$\begin{aligned} & \Pr[\exists 0 \neq v \in \ker T \cap \mathbb{Z}^m : \|v\|_2 \leq R] \\ & \leq 28^m \left(\frac{\sqrt{2}}{W}\right)^k \exp\left(\frac{k(m-1)^2 R}{W}\right) (1 + m^{-m/2} R^d). \end{aligned} \quad (21)$$

Proof. The lattice $\Lambda_v = v^\perp \cap \mathbb{Z}^m$ has covolume r_v by (19), since v is primitive. Choose j with $v_j \neq 0$. The $m-1$ independent kernel vectors $v_j e_i - v_i e_j$, $i \neq j$, have length at most r_v . Rounding coefficients in their parallelepiped shows that the covering radius of Λ_v is at most $\mu = (m-1)r_v/2$; they need not generate the full lattice.

The Euclidean Voronoi cell of the full lattice has volume r_v and radius at most μ . Its translates centered at lattice points in $[-H, H]^m$ have disjoint interiors and lie in $v^\perp \cap [-H - \mu, H + \mu]^m$. Ball's cube-slicing theorem [20, 21] bounds that section's volume by $\sqrt{2}(2H + (m-1)r_v)^{m-1}$. Divide by $r_v W^m$ and use $1 + t \leq e^t$ to obtain (20).

Let $N(t)$ count nonzero integer vectors of Euclidean norm at most t . Disjoint unit cubes centered at such vectors fit in the ball of radius $t + \sqrt{m}/2$. The standard bound $\text{vol}(B_2^m) \leq (\sqrt{2\pi e}/\sqrt{m})^m$ gives

$$N(t) \leq 8^m \quad (1 \leq t \leq \sqrt{m}), \quad N(t) \leq 8^m m^{-m/2} t^m \quad (t \geq \sqrt{m}).$$

The two ranges use respectively $t + \sqrt{m}/2 \leq 3\sqrt{m}/2$ and $t + \sqrt{m}/2 \leq 3t/2$, together with $(3/2)\sqrt{2\pi e} < 8$. Partial summation, including the possible atom at norm one, yields

$$\begin{aligned} \sum_{0 < \|v\|_2 \leq R} \|v\|_2^{-k} &= N(R)R^{-k} + k \int_1^R N(t)t^{-k-1} dt \\ &\leq 8^m \left[1 + (1 + k/d)m^{-m/2}R^d \right] \leq 2 \cdot 8^m \left(1 + m^{-m/2}R^d \right). \end{aligned}$$

Here $k \leq d$. Every nonzero integer kernel vector has a primitive divisor in the kernel. Raise (20) to the k th power for the independent rows, then sum over primitive vectors, overcounting them by the displayed sum. This proves (21). $\square$

Lemma 4.3 (Quantitative short lifts). *Let n, k, m, H be integers satisfying $n \geq 16$, $1 \leq k \leq n/16$, $15n/16 \leq m \leq n$, and $H \geq 2^{20n}$. Let T be a $k \times m$ matrix with independent uniform integer entries in $[-H, H]$. Apart from an event of probability at most*

$$2^{6n}H^{-3/7} + 2n^2/H, \tag{22}$$

the following implication holds: if $T\mathbb{Z}^m = \mathbb{Z}^k$, then every $u \in \{-1, 0, 1\}^k$ has an integer lift with

$$Tw = u, \quad \|w\|_1 \leq 2^n n^{3/2} \sqrt{H}. \tag{23}$$

Proof. Put $W = 2H + 1$, $d = m - k \geq 14k$, and

$$\beta = \frac{k - 1/2}{d - 1}, \quad b = m^{m/(2(d-1))}, \quad R = \lfloor bH^\beta \rfloor.$$

The hypotheses give

$$\frac{1}{2(n-2)} \leq \beta < \frac{1}{14}, \quad k - \beta d > \frac{3}{7}, \quad \sqrt{m} \leq b \leq m, \quad R \geq bH^\beta/2 \geq \sqrt{m}.$$

For the last inequalities, $H^\beta \geq 2^{10}$ and $\lfloor x \rfloor \geq x/2$ for $x \geq 2$ justify the floor in R . Using $\log_2 n \leq n/4$ for $n \geq 16$, we also have

$$\frac{k(m-1)^2 R}{W} \leq \frac{n^4}{2H^{13/14}} \leq 1.$$

Since $m^{-m/2}b^d = b \leq n$, (21) bounds the short-vector probability by

$$2e \cdot 8^n 2^{n/2} (1+n) H^{-3/7} \leq 2^{6n} H^{-3/7}.$$

On integer surjectivity, $\Lambda = \ker T \cap \mathbb{Z}^m$ is a rank- d lattice. Equation (19) and Hadamard's inequality give $\det \Lambda \leq m^{k/2} H^k$.

Outside the short-vector event, the first successive minimum exceeds R . Minkowski's theorem (16) and $\text{vol}(B_2^d) \geq (2/\sqrt{d})^d$ yield

$$\lambda_d(\Lambda) \leq \frac{d^{d/2} m^{k/2} H^k}{R^{d-1}} \leq 2^{d-1} (d/m)^{d/2} \sqrt{H} \leq 2^n \sqrt{H}.$$

The middle inequality uses the exact cancellation $b^{d-1} = m^{m/2}$.

For rows t_i of T , put $r_i = \text{dist}(t_i, \text{span}\{t_j : j \neq i\})$. Condition on the other rows and choose a unit normal v to their span. Some $|v_j| \geq 1/\sqrt{m}$. The event $r_i < \sqrt{k}$ forces $|v \cdot t_i| < \sqrt{k}$, which, after fixing the other entries of t_i , allows at most $2\sqrt{km} + 1$ integer pivot values. Consequently

$$\Pr[\text{some } r_i < \sqrt{k}] \leq \frac{k(2\sqrt{km} + 1)}{W} \leq \frac{2n^2}{H}.$$

On the complementary event the rows are independent, and the Schur-complement identity

$$\text{tr}((TT^\top)^{-1}) = \sum_i r_i^{-2} \leq 1$$

implies $\sigma_{\min}(T) \geq 1$.

For $\|u\|_2 \leq \sqrt{k}$, the minimum-norm real solution has norm at most $\sqrt{k}$. Subtracting a lattice vector from any integer solution puts it within $d\lambda_d/2$ of that real solution: choose d independent kernel vectors of lengths at most λ_d and round coefficients in their parallelepiped. A basis of the full lattice is not required. Thus

$$\|w\|_1 \leq \sqrt{m}(\sqrt{k} + d\lambda_d/2) \leq n + 2^{n-1}n^{3/2}\sqrt{H} \leq 2^n n^{3/2}\sqrt{H}.$$

Combining the exceptional events proves the lemma. $\square$

For a fixed $w \in \mathbb{Z}^m$ and a uniform on $[-H, H]^m \cap \mathbb{Z}^m$,

$$d_{\text{TV}}(a, a + w) = 1 - \prod_j \left(1 - \frac{|w_j|}{2H + 1}\right)_+ \leq \frac{\|w\|_1}{2H + 1}. \quad (24)$$

Here d_{TV} is total-variation distance between probability laws, $d_{\text{TV}}(\mu, \nu) = \frac{1}{2} \sum_x |\mu(x) - \nu(x)|$, and $(t)_+ = \max\{t, 0\}$. The product counts the fraction of box points in the overlap of the two translates; the inequality follows from $1 - \prod_j (1 - a_j) \leq \sum_j a_j$ for $a_j \in [0, 1]$. Thus the lifts hide bounded shifts at equal factor widths.

4.2 Uniformity over sparse observations

Put $\delta = 1/16$, let $h(t) = -t \log t - (1 - t) \log(1 - t)$, and define

$$c_* = (1 - \delta) \log 2 - 2h(\delta/2) > 0, \quad \eta_n^Z = 3e^{-c_* n}, \quad H_n = 2^{100n}.$$

Numerically, $c_* > 0.3717$. We use the convention $0 \log 0 = 0$. A random factor B has independent uniform entries in $[-H, H] \cap \mathbb{Z}$.

Lemma 4.4 (Explicit good factors). *For every $n \geq 16$ and integer $H \geq H_n$, outside a set of B of probability at most $\eta_n^Z + 2^{-33n}$, every map*

$$T_{I,J} = B[I^c, J]^\top : \mathbb{Z}^{n-u} \longrightarrow \mathbb{Z}^v, \quad u = |I|, \quad v = |J| \geq 1, \quad u + v \leq \delta n,$$

is onto and every $d \in \{-1, 0, 1\}^v$ has a preimage with ℓ_1 norm at most $\rho_n H$, where $\rho_n = 2^{-48n}$. Here $I^c = [n] \setminus I$. The corresponding statement holds for an independent factor A with the same entry law and maps $A[J, I^c]$, with the same probability bound.

Proof. Each pattern meets the hypotheses of Lemmas 4.1 and 4.3, with $k = v$, $m = n - u$, and $d_0 = n - u - v \geq 15n/16$. The leading arithmetic terms sum to at most

$$3 \cdot 2^{-n} \sum_{r \leq \delta n} \binom{2n}{r} 2^r \leq 3 \exp\{-n[(1 - \delta) \log 2 - 2h(\delta/2)]\} = \eta_n^Z.$$

There are at most 4^n patterns. Their geometric remainders sum to

$$2^{8n} H^{-3/7} + 2^{2n+1} n^2 / H \leq 2^{-34n}.$$

Indeed their binary logarithms are at most $-(244/7)n$ and $-97n$. For the three arithmetic remainders in (17), including the factor 4^n , binary logarithms are bounded respectively by

$$-97n, \quad 3n + 1 - \frac{2985}{32} n^2, \quad 3n + 1 + \log_2 n - 100n(13n/16 + 1).$$

Here $\log_2(4n) \leq n/2$, $d_0 \geq 15n/16$, and $m - 2k + 1 \geq 13n/16 + 1$. The first arithmetic error term is at most 2^{-97n} and the other two are each at most 2^{-98n} . Their sum is therefore at most 2^{-96n} ; together with the geometric error this is below 2^{-33n} . Finally,

$$\frac{2^n n^{3/2}}{\sqrt{H}} \leq 2^{-49n} n^{3/2} \leq 2^{-48n},$$

so Lemma 4.3 supplies every required lift. All bounds decrease with H . $\square$

4.3 Row-local correction and residual transfer

Residuals in this section belong to the finite Boolean cube $\mathcal{D}_n = \{0, 1\}^{n \times n}$. This is a set of integer residuals; it does not change multiplication to Boolean or mod-two multiplication. Use the Boolean characters and space $\mathcal{W}_K$ of Section 2, and put $V_L = V_L(\mathcal{X}_{n,H})$. A function has row Fourier degree at most K if every character in its Fourier expansion meets at most K rows; column Fourier degree is analogous. Thus $\mathcal{W}_K$ is the intersection of these subspaces. Operator norms in this subsection are induced by the supremum norms on the full finite domains. Define

$$(\mathcal{R}_H p)(D) = \mathbb{E}_{A,B} p(A, B, AB - D),$$

where A, B are independent inner-box factors of width H .

Theorem 4.5 (Uniform integer transfer). *For every fixed $a > 0$, there is an $n_0(a)$ such that, for every $n \geq n_0(a)$ and integer $H \geq H_n$, simultaneously for integers $0 \leq L \leq an^{5/3}$ there is a real linear map $\mathcal{H}_{H,L} : V_L \rightarrow \mathcal{W}_K$, $K = \lfloor 16L/n \rfloor$, satisfying*

$$\|\mathcal{R}_H p - \mathcal{H}_{H,L} p\|_\infty \leq e^{-c_* n/2} \|p\|_\infty. \quad (25)$$

The box threshold $H_n = 2^{100n}$ is independent of p , L , and a .

Proof. We first construct a corrected averaging operator for one row. All spaces below are finite and carry the supremum norm. Write

$$\mathcal{Z} = I_H^n \times J_{n,H}^n, \quad \mathcal{D} = \{0, 1\}^n,$$

and, for a fixed inner-box factor B , define

$$(T_B F)(d) = \mathbb{E}_{a \in [-H, H]^n \cap \mathbb{Z}^n} F(a, aB - d), \quad (S_B F)(d) = (T_B F)(0).$$

Both operators map functions on $\mathcal{Z}$ to functions on $\mathcal{D}$ and have norm one; S_B has constant output.

Suppose B is good in Lemma 4.4. If F depends only on a coordinates I and c coordinates J , with $|I| + |J| < n/16$, then

$$\|(T_B - S_B)F\|_\infty \leq \rho_n \|F\|_\infty. \quad (26)$$

Indeed, for each d , choose an integer w zero on I such that $wB[:, J] = -d_J$ and $\|w\|_1 \leq \rho_n H$. If J is empty, take $w = 0$. The observed arguments of F at $(a, aB - d)$ and $(a + w, (a + w)B)$ agree. Since $\rho_n \leq 1$, we have $|a_j + w_j| \leq 2H$ and $|(a + w)B_j| \leq 2nH^2$; also $|(aB - d)_j| \leq nH^2 + 1$. Hence both arguments lie in $\mathcal{Z}$, and (24) gives $d_{\text{TV}}(a, a + w) \leq \rho_n/2$. Apply the signed-test TV inequality to $x \mapsto F(x, xB)$ on the union of these two supports; its absolute value there is at most $\|F\|_\infty$. The inequality $|\mathbb{E}_\mu F - \mathbb{E}_\nu F| \leq 2d_{\text{TV}}(\mu, \nu) \|F\|_\infty$ proves (26). The translation does not depend on a or on the values of F .

Put $r = \lceil n/16 \rceil$. For each of the $2n$ coordinates of $\mathcal{Z}$, let Z_t set that coordinate to zero. Define the anchored local projection

$$Q_S = \prod_{t \in S} (I - Z_t) \prod_{t \notin S} Z_t, \quad Q = \sum_{|S| < r} Q_S, \quad d_n = \sum_{s=0}^{r-1} \binom{2n}{s} 2^s \leq 3^{2n}.$$

These operators act on all functions on $\mathcal{Z}$, not merely on a low-degree subspace. The operators Z_t commute and have norm one. Their products satisfy $\sum_S Q_S = I$. If F depends on a coordinate set U , then $Q_S F = 0$ whenever $S \not\subseteq U$. Thus $Q_S F$ depends only on S and has norm at most $2^{|S|} \|F\|_\infty$; Q is the identity on local coordinate degree less than r . Since $|S| < r$ implies $|S| < n/16$, (26) gives

$$\|(T_B - S_B)Q\| \leq \rho_n d_n =: \epsilon_n.$$

Define the signed, not necessarily positive, operator

$$\tilde{T}_B = T_B - (T_B - S_B)Q.$$

Then

$$\|\tilde{T}_B - T_B\| \leq \epsilon_n, \quad \|\tilde{T}_B\| \leq 1 + \epsilon_n, \quad \tilde{T}_B F = S_B F \quad (\deg F < r). \quad (27)$$

For fixed B , regard $p_B = p(A, B, C)$ as a function on $\mathcal{Z}^n$. It has norm at most $\|p\|_\infty$ and coordinate degree at most L . Conditional residual averaging is exactly $T_B^{\otimes n} p_B$. The corrected image $\tilde{T}_B^{\otimes n} p_B$ has row Fourier degree at most $\lfloor L/r \rfloor \leq K$. To verify this, expand algebraically in assignment indicators on at most L coordinates. Such an indicator factors across rows. If it specifies $s_i < r$ coordinates in row i , (27) makes that output row constant. Since $\sum_i s_i \leq L$, at most $\lfloor L/r \rfloor$ rows remain. This expansion identifies the output subspace only; no bound on the expansion's coefficients is used.

Supremum operator norms on finite spaces are maximum absolute row sums and therefore multiply under tensor products, even for signed operators. Telescoping (27) therefore gives

$$\|\tilde{T}_B^{\otimes n} - T_B^{\otimes n}\| \leq \epsilon_n \sum_{j=0}^{n-1} (1 + \epsilon_n)^j = (1 + \epsilon_n)^n - 1. \quad (28)$$

Set

$$q_R(D) = \mathbb{E}_B \mathbf{1}[B \text{ good}] (\tilde{T}_B^{\otimes n} p_B)(D).$$

The integrand is defined to be zero for bad B . This average is linear in p and has row Fourier degree at most K . With $g = \mathcal{R}_{HP}$, discarding bad B on the original norm-one averaging operator costs only their probability. Hence

$$\|g - q_R\|_\infty \leq (\eta_n^Z + 2^{-33n} + (1 + \epsilon_n)^n - 1) \|p\|_\infty. \quad (29)$$

Here $\epsilon_n \leq 3^{2n} 2^{-48n} \leq 2^{-44n}$. For $n \geq 16$,

$$(1 + \epsilon_n)^n - 1 \leq e^{n\epsilon_n} - 1 \leq 2n\epsilon_n \leq 2^{-43n}.$$

Since $c_* < \log 2$, the coefficient in (29) is at most $2\eta_n^Z$. Apply the same construction separately to $(B^\top, A^\top, C^\top)$, fixing good A . This gives q_C of column Fourier degree at most K with the same bound. Neither construction conditions both factors to be good.

Use the row projection P_K from the proof of Theorem 3.3, now averaging Boolean residual rows. It has norm at most $M_{n,K}$, fixes q_R , and preserves column degree. Thus $\mathcal{H}_{H,L}p = P_K q_C \in \mathcal{W}_K$, and the same triangle estimate, using both one-sided errors $2\eta_n^Z \|p\|_\infty$, gives

$$\|g - \mathcal{H}_{H,L}p\|_\infty \leq 2\eta_n^Z(1 + 2M_{n,K}) \|p\|_\infty. \quad (30)$$

As there, $\log M_{n,K} = O_a(n^{2/3} \log n) = o(n)$ uniformly for $L \leq an^{5/3}$, including $K = 0$. Since $\eta_n^Z = 3e^{-c_*n}$, this proves (25) after increasing $n_0(a)$. All operations defining $\mathcal{H}_{H,L}$ are real linear. $\square$

4.4 Completing the integer lower bound

Proof of Theorem 1.3. For each Boolean residual D , let μ_D be the law of $(A, B, AB - D)$ with independent inner-box factors. These triples belong to $\mathcal{X}_{n,H}$ and satisfy $f = \mathbf{1}[D = 0]$. Theorem 4.5 verifies Definition 2.3, uniformly over $H \geq H_n$. Proposition 2.7 supplies the degree and query lower bounds; the stored-submatrix verifier supplies the matching upper bound. At $H = H_n$, every entry in the outer domain has magnitude at most $4n2^{200n} + 1$, hence has $O(n)$ bits. The oracle returns that complete exact value, so this statement is not a bit-query lower bound. $\square$

5 Shared distributional and adversary lower bounds

We prove Theorem 1.4 for both arithmetic settings at once. Let Ω be either $\Omega_{n,q}$ or $\mathcal{X}_{n,H}$, and $f = \mathbf{1}[AB = C]$ on that finite product domain. Use the corresponding Boolean residual averaging map and transfer operator established in Sections 3 and 4. The argument below uses only Definition 2.3. In particular, its constants are uniform in q and in $H \geq H_n$, respectively.

5.1 A quantitative collision test

Lemma 5.1 (A quantitative collision test). *Suppose $h \in \mathcal{W}_K$, $0 \leq h \leq 1$, and*

$$\alpha = h(0), \quad \beta = \frac{1}{n^2} \sum_{i,j} h(E_{ij}), \quad \alpha - \beta \geq 1/32.$$

Then $K \geq c_0 n^{2/3}$ for all sufficiently large n , where $c_0 > 0$ is an absolute constant.

Proof. Set $N = 32n^2$ and $M = 48n^2$. Given a source map $u : [N] \rightarrow [M]$, choose $2n$ distinct domain positions uniformly in order, put the first n into a list I and the last n into a list J , and set

$$\Psi(u) = \mathbb{E}_{I,J} h(D(u|_I, u|_J)).$$

For each choice the polynomial degree is at most $2K$ by Lemma 2.5: each sampled label is replaced by the value of u at its fixed sampled position. This substitution sends each one-hot label indicator to a single one-hot source indicator. The sampling law is independent of u , so its average has the same degree bound. Equal *values* at different positions do not identify the source coordinates. Also

$0 \leq \Psi \leq 1$ for every source map, whether or not it satisfies the collision promise. If u is injective, $\Psi(u) = \alpha$.

Now fix any exactly two-to-one map u . Its equal-value pairs form a perfect matching on the N domain positions. Let Z count matched pairs with one endpoint in each sampled list. Equivalently, Z is the number of nonzero entries of $D(u|_I, u|_J)$. Writing Z as the sum of its n^2 cross-pair indicators gives

$$\mu := \mathbb{E}Z = \frac{n^2}{N-1}, \quad \mu_2 := \mathbb{E}Z(Z-1) = \frac{n^2(n-1)^2}{(N-1)(N-3)}. \quad (31)$$

For the first identity, each of the n^2 cross pairs is a matching edge with probability $1/(N-1)$. For the second, two matching edges cannot share an endpoint. There are $n^2(n-1)^2$ ordered pairs of disjoint cross edges, and the probability that both belong to a uniform perfect matching is $1/((N-1)(N-3))$. Thus

$$\mu \geq 1/32, \quad \frac{\mu_2}{\mu} = \frac{(n-1)^2}{32n^2-3} \leq 1/32. \quad (32)$$

Put $P_1 = \Pr[Z = 1]$ and $P_{\geq 2} = \Pr[Z \geq 2]$. Since $z \leq z(z-1)$ for integers $z \geq 2$,

$$P_1 \geq \mu - \mu_2, \quad P_{\geq 2} \leq \mu_2/2.$$

On $Z = 0$ the residual is zero. Conditional on $Z = 1$, the sole nonzero position is uniform in $[n]^2$: independent permutations of the two list orders act transitively on these positions and preserve the sampling law and the event. Hence the conditional average of h is β . On $Z \geq 2$ it is some $w \in [0, 1]$ (take any such w if this event is null). It follows that

$$\begin{aligned} \alpha - \Psi(u) &= P_1(\alpha - \beta) + P_{\geq 2}(\alpha - w) \\ &\geq (\mu - \mu_2)(\alpha - \beta) - \mu_2/2 \\ &\geq \mu \left(\frac{31}{32} \frac{1}{32} - \frac{1}{64} \right) \geq \frac{15}{32768} =: \gamma. \end{aligned} \quad (33)$$

To amplify, put $\theta = \alpha - \gamma/2$ and choose an integer $r \geq 2\gamma^{-2} \log 6$. The polynomial

$$A_\alpha(t) = \Pr[\text{Bin}(r, t)/r \geq \theta]$$

is the sum of the binomial terms with $j \geq \lceil r\theta \rceil$. The hypothesis gives $\alpha \geq 1/32$, while $\gamma < 1/32$, so $0 < \theta < 1$. Hence A_α has degree at most r and lies in $[0, 1]$ throughout $[0, 1]$. Hoeffding's inequality [11] gives $A_\alpha(\alpha) \geq 5/6$ and $A_\alpha(t) \leq 1/6$ for $0 \leq t \leq \alpha - \gamma$, since both tail events are at distance at least $\gamma/2$ from the mean. Thus $A_\alpha(\Psi)$ is a collision approximant of degree at most $2rK$, globally bounded on all source maps. The imported lower bound yields $2rK \geq c_{\text{col}}N^{1/3}$. Taking $c_0 = c_{\text{col}}32^{1/3}/(2r)$ proves the lemma; r is an absolute constant independent of α and h . $\square$

5.2 The uniform seed and average-case hardness

Let $\mathcal{A}$ be $\mathbb{F}_q^{n \times n}$ in the finite-field case and $([-H, H] \cap \mathbb{Z})^{n \times n}$ in the integer case, and put $b = |\mathcal{A}|$. Let P_{yes} be the law of (A, B, AB) for independent uniform $A, B \in \mathcal{A}$, and let P_{no} be the law of $(A, B, AB - E_{ij})$ with (i, j) independently uniform. Define

$$\lambda_0 = \frac{1}{2}(P_{\text{yes}} - P_{\text{no}}). \quad (34)$$

For a signed measure ρ on Ω , write $\|\rho\|_1 = \sum_{z \in \Omega} |\rho(z)|$ and $\rho(p) = \sum_z \rho(z)p(z)$. Explicitly,

$$\lambda_0(A, B, C) = \begin{cases} (2b^2)^{-1}, & (A, B) \in \mathcal{A}^2, C = AB, \\ -(2n^2b^2)^{-1}, & (A, B) \in \mathcal{A}^2, AB - C = E_{ij} \text{ for some } i, j, \\ 0, & \text{otherwise.} \end{cases} \quad (35)$$

The two supports are disjoint, so $\|\lambda_0\|_1 = 1$, $\lambda_0(1) = 0$, and $\lambda_0(f) = 1/2$. Although the integer seed is supported on inner-box factors, all function norms and all LP variables below use the full outer domain $\mathcal{X}_{n,H}$.

Theorem 5.2 (Restricted norm bound). *There are $a > 0$ and n_0 such that, for all $n \geq n_0$ and integers $0 \leq L \leq an^{5/3}$,*

$$\sup\{|\lambda_0(p)| : p \in V_L, \|p\|_\infty \leq 1\} \leq 1/16. \quad (36)$$

Proof. Choose $a = \min\{1, c_0/32\}$, where c_0 is from Lemma 5.1. Suppose (36) fails. Replace p by $-p$ if necessary, so that $\lambda_0(p) > 1/16$, and put $p_+ = (1 + p)/2$. Then $0 \leq p_+ \leq 1$ on all triples. Its residual average $g = \mathcal{R}p_+$ satisfies the exact identity

$$g(0) - \frac{1}{n^2} \sum_{i,j} g(E_{ij}) = \lambda_0(p) > 1/16.$$

Indeed, the left side is $\mathbb{E}_{P_{\text{yes}}}p_+ - \mathbb{E}_{P_{\text{no}}}p_+ = (\mathbb{E}_{P_{\text{yes}}}p - \mathbb{E}_{P_{\text{no}}}p)/2$. Apply Corollary 2.4, with $K = \lfloor 16L/n \rfloor$. Uniformly over the indicated range of L , $\xi_{n,L} \leq 1/100$ for all sufficiently large n . The resulting $h \in \mathcal{W}_K$ is bounded in $[0, 1]$ and obeys

$$h(0) - \frac{1}{n^2} \sum_{i,j} h(E_{ij}) \geq \frac{1/16 - 2\xi_{n,L}}{1 + 2\xi_{n,L}} \geq \frac{1}{24} > \frac{1}{32}.$$

Lemma 5.1 forces $K \geq c_0n^{2/3}$. On the other hand, $K \leq 16an^{2/3} \leq (c_0/2)n^{2/3}$, a contradiction. Increase n_0 to cover the large- n requirements of both lemmas. $\square$

Corollary 5.3 (Average-case hardness). *The distributional assertion of Theorem 1.4 holds.*

Proof. Let p be the probability that the algorithm outputs *equal*, and set $a_1 = \mathbb{E}_{P_{\text{yes}}}p$ and $b_1 = \mathbb{E}_{P_{\text{no}}}p$. Under the balanced distribution its success probability is $(a_1 + 1 - b_1)/2$, so success at least $2/3$ gives $a_1 - b_1 \geq 1/3$. The acceptance function is bounded on all inputs and has degree at most twice the worst-case query count T , by Lemma 2.2, including its finite-history argument for adaptive algorithms. Thus $s = 2p - 1 \in V_{2T}$ has $\|s\|_\infty \leq 1$ and

$$\lambda_0(s) = a_1 - b_1 \geq 1/3,$$

contradicting Theorem 5.2 if $2T \leq an^{5/3}$. This proves $T = \Omega(n^{5/3})$ without amplifying average success pointwise. $\square$

5.3 An exact finite dual witness

We use the assignment notation of Section 2; the empty assignment is included throughout.

For $0 \leq \ell \leq 3n^2$, consider the finite linear program

$$\begin{aligned} \tau_\ell = \min_{u_z, v_z \geq 0} \quad & \sum_{z \in \Omega} (u_z + v_z) \\ \text{subject to} \quad & \sum_{z \sim \alpha} (u_z - v_z) = \sum_{z \sim \alpha} \lambda_0(z) \quad \text{for every } |\alpha| \leq \ell. \end{aligned} \quad (37)$$

Lemma 5.4 (Exact norm extension). *The optimum is attained, and*

$$\tau_\ell = \sup_{p \in V_\ell, \|p\|_\infty \leq 1} |\lambda_0(p)|. \quad (38)$$

Moreover $\tau_0 = 0$, $\tau_{3n^2} = 1$, and τ_ℓ is nondecreasing.

Proof. The Jordan parts of λ_0 give a feasible point of value one; the feasible objective sublevel set at one is compact, so the optimum exists. With free dual variables y_α , put $p(z) = \sum_{|\alpha| \leq \ell} y_\alpha \mathbf{1}[z \sim \alpha]$. The two dual inequalities at z are precisely $-1 \leq p(z) \leq 1$, and the objective is $\lambda_0(p)$. Lemma 2.1 and finite LP strong duality give the supremum without the absolute value. The feasible set of p is symmetric under $p \mapsto -p$, which gives (38). At degree zero the only constraint is total mass; it is satisfied by $u = v = 0$ because $\lambda_0(1) = 0$. At full degree, the constraint for the complete assignment associated with each input z forces $u_z - v_z = \lambda_0(z)$, so every feasible objective is at least $\|\lambda_0\|_1 = 1$; the Jordan decomposition attains one. Adding constraints proves monotonicity. $\square$

Define the integer

$$L_n = \max\{\ell \in \{0, \dots, 3n^2\} : \tau_\ell \leq 1/16\}. \quad (39)$$

It is well-defined and smaller than $3n^2$. Theorem 5.2 implies $L_n \geq \lfloor an^{5/3} \rfloor$ for all sufficiently large n . Thus the recipe does not need the numerical constant hidden in the imported collision lower bound as an input.

Choose an optimum of (37) for $\ell = L_n$, set $\sigma = u - v$, and define

$$\lambda = \lambda_0 - \sigma, \quad \Lambda = \|\lambda\|_1, \quad \phi = \lambda/\Lambda. \quad (40)$$

Proposition 5.5. *The signed measure ϕ satisfies*

$$\|\phi\|_1 = 1, \quad \phi(p) = 0 \ (p \in V_{L_n}), \quad \phi(f) \geq 7/17.$$

In particular $\phi(1) = 0$, and its positive and negative parts both have total mass $1/2$.

Proof. The LP gives $\|\sigma\|_1 \leq 1/16$. Its constraints say that σ and λ_0 agree on every assignment indicator of size at most L_n ; Lemma 2.1 extends this equality to all of V_{L_n} . Thus λ annihilates that space, and

$$\Lambda \leq 17/16, \quad \lambda(f) \geq 1/2 - 1/16 = 7/16.$$

In particular $\Lambda > 0$, and normalization gives the correlation bound. The empty-assignment constraint gives total mass zero, from which the final assertion follows. $\square$

5.4 The negative-weight adversary matrix

We specify the adversary, check the assignment-isometry conversion [5], and deduce the general-input adversary bound.

For disjoint sets X of YES inputs and Y of NO inputs, the rectangular negative-weight adversary value is

$$\text{Adv}^\pm(X, Y) = \sup_{\Gamma} \{ \|\Gamma\| : \|\Gamma \circ \Delta_t\| \leq 1 \text{ for every } t \}, \quad (\Delta_t)_{x,y} = \mathbf{1}[x_t \neq y_t].$$

Here Γ is real, has rows X and columns Y , $\circ$ is the entrywise product, and norms are Euclidean operator norms. For a total function we write $\text{Adv}^\pm(f) = \text{Adv}^\pm(f^{-1}(1), f^{-1}(0))$. The general adversary theorem gives bounded-error query complexity $\Omega(\text{Adv}^\pm(X, Y))$ for partial functions over

finite input alphabets [10, 14]; the nonbinary filtering convention used below is also spelled out in [5, Proposition 4]. This agrees with the symmetric formulation by putting Γ and $\Gamma^\top$ in opposite off-diagonal blocks; the operator norm and all filtered norms are unchanged. The lower-bound argument also allows arbitrary Hilbert-space workspace: its progress function uses only finitely many input-pair inner products, and its per-query estimate uses $\|M \otimes I_{\mathcal{H}}\| = \|M\|$ for finite matrices M . Together with the finite-label simulation in Section 1.1, this justifies its use on finite restrictions of an infinite ring.

Set $\pi_+ = 2\phi^+$ and $\pi_- = 2\phi^-$. These are probability measures on Ω . For every assignment α of size at most L_n , the identity $\phi(\mathbf{1}[\cdot \sim \alpha]) = 0$ gives $\pi_+(\{z : z \sim \alpha\}) = \pi_-(\{z : z \sim \alpha\})$. Put $m = \lfloor L_n/2 \rfloor$; assume n is large enough that $m \geq 1$. For $0 \leq k \leq m$ define matrices with rows indexed by $z \in \Omega$ and columns by assignments of size at most k :

$$(V_k^\pm)_{z,\alpha} = \sqrt{\pi_\pm(z)} \mathbf{1}[z \sim \alpha]. \quad (41)$$

Products of two assignment indicators vanish when inconsistent and otherwise are assignment indicators of size at most $2k \leq L_n$. Consequently

$$G_k = (V_k^+)^* V_k^+ = (V_k^-)^* V_k^-.$$

Use the Moore–Penrose pseudoinverse to set

$$W_k = V_k^+ G_k^\dagger (V_k^-)^*, \quad \tilde{\Gamma} = \sum_{k=0}^{m-1} W_k. \quad (42)$$

All these matrices are real; the adjoint notation is retained for convenience. The positive and negative spaces are two copies of $\mathbb{R}^\Omega$. Inputs with zero π_+ mass give zero rows, and inputs with zero π_- mass give zero columns, so no division by a possibly zero probability is involved.

Proposition 5.6 (Adversary conversion bounds). *The matrix in (42) satisfies*

$$\|\tilde{\Gamma}\| = m, \quad \tilde{\Gamma} \sqrt{\pi_-} = m \sqrt{\pi_+}, \quad \|\tilde{\Gamma} \circ \Delta_t\| \leq 2 \quad \text{for every entry coordinate } t.$$

For subsets $X, Y \subseteq \Omega$ with $\vartheta = \pi_+(X) - \pi_-(\Omega \setminus Y) > 0$,

$$\|\tilde{\Gamma}[X, Y]\| \geq m\vartheta/2.$$

Here brackets denote restriction to the indicated rows and columns.

Proof. Apply the construction in [5, Theorem 15], together with Proposition 4 there for nonbinary alphabets; all numbering here refers to the published TQC version. Its hypotheses are equality of assignment probabilities through size $2m$ on two finite-alphabet probability spaces. Here they hold because $\pi_\pm = 2\phi^\pm$ are probability measures and $2m \leq L_n$. The possibly different coordinate alphabets may be embedded in one finite label set; coordinate inequalities and assignment moments are unchanged.

The common-Gram identity above identifies our pseudoinverse formula with the source’s assignment isometry:

$$W_k V_k^- c = V_k^+ G_k^\dagger G_k c = V_k^+ c, \quad W_k|_{(\text{ran } V_k^-)^\perp} = 0.$$

Indeed, $\ker V_k^+ = \ker V_k^- = \ker G_k$, and $G_k^\dagger G_k$ is the orthogonal projection onto $(\ker G_k)^\perp$. Thus the displayed identity holds even when G_k is singular. It also shows that W_k is an isometry from

$\text{ran } V_k^-$ onto $\text{ran } V_k^+$ and is zero on the orthogonal complement. Thus each W_k is a partial isometry, and $\|W_k\| \leq 1$. For the empty assignment $\emptyset$ one has $V_k^\pm e_\emptyset = \sqrt{\pi_\pm}$, so $W_k \sqrt{\pi_-} = \sqrt{\pi_+}$ for every k ; summing over k gives $\tilde{\Gamma} \sqrt{\pi_-} = m \sqrt{\pi_+}$. The triangle inequality gives $\|\tilde{\Gamma}\| \leq \sum_{k=0}^{m-1} \|W_k\| \leq m$, whereas the displayed unit-vector identity gives the reverse inequality. Hence $\|\tilde{\Gamma}\| = m$. These isometries are compatible: on $\text{ran } V_k^-$, both W_k and W_m send each assignment vector to the corresponding positive vector. Thus W_k is the restriction of the common isometry W_m , extended by zero. Spans of assignments of size at most k equal those of size exactly k by summing extensions on unused coordinates (there are enough because $k \leq m < 3n^2$); thus both conventions give the same construction. Zero-probability rows and columns may be omitted or padded with zeros.

This recovers Theorem 15(a) of [5]. For each entry, Theorem 15(b), equation (16), gives a norm-one matrix with the same disagreement-filtered entries, using the same summation range $0 \leq k \leq m-1$. Proposition 4 yields the bound two, not the Boolean bound one.

For completeness, set $a = \pi_+(X)$ and $b = \pi_-(\Omega \setminus Y)$, so $a-b = \vartheta > 0$. The restriction calculation is

$$\|\tilde{\Gamma}[X, Y]\| \geq m(\sqrt{a} - \sqrt{b}) \geq m\vartheta/2.$$

Indeed, let $u = \sqrt{\pi_-}|_Y$. Since $b < 1$, this vector is nonzero, and $\|u\| \leq 1$. Restricting $\tilde{\Gamma} \sqrt{\pi_-} = m \sqrt{\pi_+}$ to the rows in X gives

$$\begin{aligned} \|\tilde{\Gamma}[X, Y]u\| &\geq m \|\sqrt{\pi_+}|_X\| - \|\tilde{\Gamma}[X, \Omega \setminus Y](\sqrt{\pi_-}|_{\Omega \setminus Y})\| \\ &\geq m\sqrt{a} - m\sqrt{b}. \end{aligned}$$

Dividing by $\|u\| \leq 1$ proves the first inequality. The second is $\sqrt{a} - \sqrt{b} = (a-b)/(\sqrt{a} + \sqrt{b}) \geq (a-b)/2$. This is the restriction step of [5, proof of Theorem 17], applied here to the finite-alphabet matrix, and completes the specialization. $\square$

5.5 The singleton-residual strengthening

Proof of the adversary and promise assertions in Theorem 1.4. Set $X = f^{-1}(1)$, $Y = f^{-1}(0)$, and $Y_1 = \{AB - C = E_{ij} \text{ for some } i, j\} \subseteq Y$. The correction σ has total mass zero by the empty-assignment constraint and satisfies $\|\sigma\|_1 \leq \kappa = 1/16$; hence its positive and negative masses are each at most $\kappa/2$. The seed has mass $1/2$ on X , is nonnegative there and on $\Omega \setminus Y_1$, and is nonpositive on Y_1 . Consequently

$$\begin{aligned} \lambda^+(X) &\geq \lambda(X) = 1/2 - \sigma(X) \geq 1/2 - \kappa/2, \\ \lambda^-(\Omega \setminus Y_1) &\leq \sigma^+(\Omega \setminus Y_1) \leq \kappa/2. \end{aligned}$$

Using $\pi_\pm = 2\lambda^\pm/\Lambda$ and $0 < \Lambda \leq 1 + \kappa$ gives

$$\pi_+(X) - \pi_-(\Omega \setminus Y_1) = \frac{2}{\Lambda}(\lambda^+(X) - \lambda^-(\Omega \setminus Y_1)) \geq \frac{1-2\kappa}{1+\kappa} = \frac{14}{17}.$$

Proposition 5.6 therefore makes $\Gamma_1 = \frac{1}{2}\tilde{\Gamma}[X, Y_1]$ a valid adversary: the nonbinary filtered-norm bound two becomes one after scaling, and $\|\Gamma_1\| \geq 7m/34$. The correction need not be supported on the promise. Enlarging the NO set to all nonzero residuals supported on one entry permits zero-column padding, which preserves the operator norm and every filtered norm.

For general inputs define

$$\Gamma = \frac{1}{2}\tilde{\Gamma}[X, Y]. \tag{43}$$

Its filtered norms are at most one by the same proposition, and $\Gamma[X, Y_1] = \Gamma_1$. Since restriction cannot increase operator norm,

$$\text{Adv}^\pm(f) \geq \|\Gamma\| \geq \|\Gamma_1\| \geq \frac{7m}{34} = \frac{7}{34} \left\lfloor \frac{L_n}{2} \right\rfloor = \Omega(n^{5/3}). \quad (44)$$

Equations (35), (37), (39), (40), (41), (42), and (43), together with restriction to Y_1 , specify every matrix used. The adversary theorem supplies the corresponding query lower bounds. This proves the general-input and promise adversary assertions; the distributional part was proved in Corollary 5.3. $\square$

6 Extension to unital rings

We combine the arithmetic branches by transferring finite restrictions to R , which may be infinite, noncommutative, or have zero divisors. No infinite-dimensional adversary theorem is needed.

Proof of Theorem 1.1. Characteristic-zero lower bound. If $\text{char } R = 0$, the map $\iota : \mathbb{Z} \rightarrow R$ given by $\iota(a) = a1_R$ is an injective ring homomorphism with central image. Apply it to every entry of an integer instance. Then

$$\iota(A)\iota(B) - \iota(C) = \iota(AB - C),$$

so verification is preserved exactly. In particular, the finite hard restriction of Theorems 1.3 and 1.4 embeds in R without changing its matrix dimension.

Positive-characteristic lower bound. Let $\text{char } R = m \geq 2$, choose a prime $p \mid m$, and set

$$e = (m/p)1_R.$$

The additive order of 1_R is m , so e has additive order exactly p . For matrices A, B, C over $\mathbb{F}_p$, let $\hat{A}, \hat{B}, \hat{C}$ be their entrywise integer representatives in $\{0, \dots, p-1\}$. Define

$$A'_{ij} = \hat{A}_{ij}1_R, \quad B'_{ij} = \hat{B}_{ij}e, \quad C'_{ij} = \hat{C}_{ij}e. \quad (45)$$

Distributivity and multiplication by integer multiples of the identity give

$$(A'B' - C')_{ij} = \left(\sum_k \hat{A}_{ik} \hat{B}_{kj} - \hat{C}_{ij} \right) e. \quad (46)$$

Since an integer multiple te vanishes exactly when $p \mid t$, $A'B' = C'$ in R if and only if $AB = C$ in $\mathbb{F}_p$. We therefore apply the finite-prime-field query and adversary lower bounds of Theorems 1.2 and 1.4. No embedded field in R is required. The encoding is intentionally asymmetric: e need not be idempotent, so multiplying both factors by e would not give (46).

Oracle and adversary preservation. Each transformed entry depends on one source entry through a fixed map. One can simulate its value oracle by querying that entry into an ancilla, computing the encoded answer reversibly, and uncomputing the ancilla, using a constant number of source queries. Moreover, all coordinate maps are injective on their source alphabets. This is immediate in characteristic zero. In (45), the map $a \mapsto a1_R$ is injective on $\{0, \dots, p-1\}$ because $p \leq m$, and $a \mapsto ae$ is injective because e has order p . Writing Φ for either triple encoding, for every coordinate t we have

$$\mathbf{1}[\Phi(x)_t \neq \Phi(y)_t] = \mathbf{1}[x_t \neq y_t].$$

Relabel the rows and columns of a source adversary matrix by Φ . Its operator norm and every filtered norm remain identical. Thus the finite adversary witnesses themselves transfer, not merely the query lower bounds.

Upper bound. Use the stored-submatrix variant *Verify Full* of Buhrman and Špalek [6, Sections 4.1 and 6.1]. More generally, consider $A \in R^{t \times k}$, $B \in R^{k \times t}$, and $C \in R^{t \times t}$ with $1 \leq t \leq 2k$. For $t \geq 8$, choose independent uniform r -subsets $U, V \subseteq [t]$, where $1 \leq r \leq t/2$, and cache

$$A[U, :], \quad B[:, V], \quad C[U, V].$$

Preparing the subset superposition and these data costs $S = O(kr + r^2) = O(kr)$ queries. A lazy product Johnson walk changes one selected row or column at a time; updating the cache and uncomputing removed entries costs $U_{\text{qry}} = O(k + r) = O(k)$ queries. The Johnson-chain gap is $t/(r(t - r))$, so laziness and updating one coordinate give product-chain gap $\delta = \Theta(1/r)$. A state is marked when $A[U, :]B[:, V] \neq C[U, V]$. This exact check uses only cached values and ordered sums, with no division or random scalar compression; the integral-domain hypothesis is therefore unnecessary. If $AB \neq C$, a fixed wrong entry is captured with probability $\varepsilon \geq (r/t)^2$; if $AB = C$, no state is marked. The quantum-walk bound [15, Theorem 3], with checking cost zero, gives bounded-error query cost

$$O\left(S + \frac{U_{\text{qry}}}{\sqrt{\delta\varepsilon}}\right) = O\left(kr + \frac{kt}{\sqrt{r}}\right). \quad (47)$$

Taking $r = \lfloor t^{2/3} \rfloor \leq t/2$ gives $O(kt^{2/3})$ queries. For $t < 8$, reading all $2kt + t^2$ entries costs $O(k)$ and suffices. Setting $t = k = n$ proves the $O(n^{5/3})$ upper bound. This is a query bound only: the exact cached checks may use substantially more arithmetic time and workspace. $\square$

In particular, Theorem 1.1 applies to every finite or infinite field, with no averaging over an infinite field required.

Singleton residuals. Here E_{ij} denotes the matrix with 1_R at (i, j) and zeros elsewhere. The characteristic-zero embedding preserves the promise residuals 0 and E_{ij} . In characteristic m , (46) instead takes a prime-field unit residual to eE_{ij} . Consequently the same lower bounds hold for the ring promise

$$AB - C \in \{0\} \cup \{eE_{ij} : i, j \in [n]\}.$$

When $m = p$ is prime, $e = 1_R$, so this is the unit-singleton promise, including for every positive-characteristic field. For composite m , this reduction alone does not establish the promise with residual $1_R E_{ij}$. For example, in $\mathbb{Z}/4\mathbb{Z}$ it uses $e = 2$ and the encoding $(A, B, C) \mapsto (\hat{A}, 2\hat{B}, 2\hat{C})$ of an $\mathbb{F}_2$ instance; $e^2 = 0$ is harmless because only B, C are scaled.

Why the hypotheses matter. Without an identity, the universal exponent fails: give a nontrivial abelian group the multiplication $xy = 0$. Then verification reduces to testing $C = 0$ and has query complexity $\Theta(n)$, by quantum search over its n^2 entries and the matching lower bound on a two-element entry restriction [4]. The zero ring has only YES inputs and needs no queries. Thus both hypotheses in Theorem 1.1 matter.

Other residual promises. Over the integers and fields, the unit-singleton theorem already implies hardness under the sole restriction that the residual have at most a specified positive number of nonzero entries, or rank at most one. More generally, any residual promise containing zero and every E_{ij} inherits the lower bound. This does not give the same conclusion when many errors are required, or when the factors themselves must be sparse or low rank. In composite characteristic, use eE_{ij} as above.

7 Rectangular verification and consequences

We extend the transfer method to rectangular products and record its localization consequence.

7.1 Rectangular products with a long inner dimension

Retaining the inner dimension while inspecting a smaller output block yields more than padding a square hard instance.

Corollary 7.1 (Square-output rectangular verification). *Let $1 \leq s \leq k$ and let R be a fixed nonzero unital associative ring. For entry-oracle matrices*

$$A \in R^{s \times k}, \quad B \in R^{k \times s}, \quad C \in R^{s \times s},$$

deciding whether $AB = C$ has bounded-error worst-case query complexity

$$Q_{1/3} = \Theta(ks^{2/3}).$$

The lower bound has finite support. Over finite fields its constants are independent of the field size; over the integers it holds on

$$\mathcal{X}_{s,k,H}^{\text{rect}} = I_H^{s \times k} \times I_H^{k \times s} \times J_{k,H}^{s \times s}, \quad H \geq H_k,$$

with absolute constants. Choosing $H = H_k = 2^{100k}$ gives $O(k)$ bits per entry. The same finite-domain lower bounds hold for bounded approximate coordinate degree.

Proof. We first prove the lower bound over a finite field and over the integer restriction in the statement. Let p be a globally $[0, 1]$ -valued degree- d approximant to the rectangular predicate. Apply the fixed majority amplifier used in the proof of Proposition 2.7. The resulting function p_* has error at most 0.01, remains globally in $[0, 1]$, and has degree at most $L = a_0 d$, where a_0 is an absolute positive integer. If $L > k^{5/3}$, then already $d > ks^{2/3}/a_0$, so assume $L \leq k^{5/3}$.

Let π project a square k -dimensional triple onto the first s rows of its first factor, the first s columns of its second factor, and the upper-left s -by- s block of its proposed product. Define $P = p_* \circ \pi$. In the integer case, π maps $\mathcal{X}_{k,H}$ onto $\mathcal{X}_{s,k,H}^{\text{rect}}$; in the field case it maps the full square product domain onto the full rectangular product domain. Consequently $P \in V_L$ and $0 \leq P \leq 1$ on the entire square domain, not just on inputs used by the residual distribution.

For $D \in \{0, 1\}^{s \times s}$ let $\iota(D)$ be the k -by- k matrix with upper-left block D and zeros elsewhere. Independence of the sampled entries gives the exact identity

$$(\mathcal{R}_k P)(\iota(D)) = \mathbb{E}_{A,B} p_*(A, B, AB - D), \quad (48)$$

where $\mathcal{R}_k$ denotes the square residual average at dimension k , and on the right A, B have the rectangular shapes in the statement. Their entries are independent and uniform in the field, or in $[-H, H] \cap \mathbb{Z}$, respectively. In the integer case all entries of $AB - D$ lie in $J_{k,H}$, so every right-hand input belongs to the stated product restriction.

Apply Theorem 3.3 or Theorem 4.5 at dimension k , and then restrict its output along ι . For the field theorem also restrict each remaining residual coordinate to $\{0, 1\}$. Each term depending on at most K rows and K columns retains this property under either restriction, so the corresponding linear span is preserved. Thus we obtain a real function $h_0 \in \mathcal{W}_K$ on $\{0, 1\}^{s \times s}$, where

$$K = \lfloor 16L/k \rfloor, \quad |h_0(D) - \mathbb{E}_{A,B} p_*(A, B, AB - D)| \leq \eta_k, \quad \eta_k = e^{-c_* k/2}.$$

For all sufficiently large k the estimate is uniform in $s \leq k$, in the field size, and in $H \geq H_k$. By (48), the averaged function is at least 0.99 at zero and at most 0.01 at every nonzero Boolean residual. For $\eta_k \leq 1/200$, normalization as in Corollary 2.4 gives

$$h = \frac{h_0 + \eta_k}{1 + 2\eta_k} \in \mathcal{W}_K, \quad 0 \leq h \leq 1, \quad h(0) \geq 0.98, \quad h(D) \leq 0.02 \quad (D \neq 0).$$

Lemma 2.6, with residual dimension s , gives $K = \Omega(s^{2/3})$. This estimate is uniform also for bounded s : separation of zero from a singleton residual forces $K \geq 1$, since $\mathcal{W}_0$ consists only of constants. Hence

$$16a_0d/k \geq K = \Omega(s^{2/3}), \quad d = \Omega(ks^{2/3}).$$

The bounded range of dimensions excluded by the transfer threshold is absorbed by reducing the absolute constant: the predicate is nonconstant and therefore has degree and bounded-error query complexity at least one. The proof of Lemma 2.2 applies to the $2sk + s^2$ addressed entries, so the coordinate-degree bound yields the query lower bound. For an arbitrary fixed R , use the characteristic-zero embedding or the asymmetric positive-characteristic encoding (45). Their residual identity (46) is valid for rectangular products as well, and each encoded query uses a constant number of source queries. Composition with either coordinatewise encoding cannot increase coordinate degree. Thus both lower bounds transfer to the corresponding finite restrictions over R .

For the upper bound, apply the stored-submatrix bound (47) with $t = s \leq k$, obtaining $O(ks^{2/3})$ queries, including the direct-reading case $s < 8$. The exact cached check applies to every ring in the statement. $\square$

This corollary settles the regime of square outputs with inner dimension at least the output side. It does not assert a sharp bound for arbitrary m -by- k times k -by- p shapes.

7.2 Locating an erroneous entry

Corollary 7.2 (Locating an error). *Over every fixed nonzero unital associative ring, the task of returning `equal` when $AB = C$, and otherwise returning an index (i, j) with $(AB)_{ij} \neq C_{ij}$, has bounded-error worst-case query complexity $\Theta(n^{5/3})$.*

Proof sketch. The lower bound follows from decision. For the upper bound, recursively verify output quadrants, increase the success probability with the depth, and check the final candidate entry exactly. The costs form a convergent geometric sum and total $O(n^{5/3})$; Appendix A.6 gives the amplification and error analysis. $\square$

Further applications. Appendix A treats fixed targets, quadratic identities, Gram comparisons, and structured inputs, including positive-definite matrices arbitrarily close to the identity. Their promises are separate.

A Applications and structural restrictions

This appendix collects applications and error-localization analysis in the exact whole-entry model of Section 1.1. None of these constructions is needed for the main lower bound or the rectangular extension.

A.1 Entry-local reductions and fixed targets

The following reductions preserve the exponent and the finite-support adversary bound. They impose different promises, which must not be combined unless the construction explicitly does so.

Reduction convention. In the constructions below, the target dimension satisfies $N = \Theta(n)$, where n is the source dimension, and each target entry is a known function of at most one source entry. A target value query therefore costs $O(1)$ source queries. Each encoding of complete inputs is injective. It also transfers the adversary lower bound. Write $M = \Gamma \circ \Delta_t$ and let P_b, Q_b project onto source rows and columns with $g(z_t) = b$. After relabeling the source matrix, the target-filtered matrix is

$$M' = M - \sum_b P_b M Q_b.$$

The row and column projections are orthogonal within their respective families, so $\|\sum_b P_b M Q_b\| \leq \|M\|$. Hence $\|M'\| \leq 2\|\Gamma \circ \Delta_t\|$, and $\Gamma/2$ is feasible. For injective coordinate maps the filters agree exactly; constant coordinates give zero filters. This observation also covers multiplication by 2 in rings where that map is not injective.

Corollary A.1 (Fixed targets and commuting inputs). *Over every fixed nonzero unital associative ring, each of the following tasks has bounded-error query complexity $\Theta(N^{5/3})$ and a finite-support negative-weight adversary lower bound $\Omega(N^{5/3})$:*

1. testing $AB = 0$ with the target matrix known;
2. testing $AB = I$, even when both factors are upper triangular, invertible, and have unit diagonal;
3. testing $A^2 = C$, even when A, C are upper triangular, invertible, have unit diagonal, and commute.

In particular, requiring all three supplied matrices to be invertible and pairwise commuting does not improve the worst-case exponent.

Proof. For an arbitrary source triple, put

$$M = \begin{pmatrix} A & -C \\ 0 & 0 \end{pmatrix}, \quad N_0 = \begin{pmatrix} B & 0 \\ I & 0 \end{pmatrix}.$$

Then $MN_0 = \text{diag}(AB - C, 0)$, proving the first reduction. For the second, use

$$U = \begin{pmatrix} I & A & C \\ 0 & I & B \\ 0 & 0 & I \end{pmatrix}, \quad V = \begin{pmatrix} I & -A & 0 \\ 0 & I & -B \\ 0 & 0 & I \end{pmatrix}.$$

Both $UV - I$ and $VU - I$ have only one potentially nonzero block, namely $C - AB$ in position $(1, 3)$. For the third, let $X_{12} = A$, $X_{23} = B$, and $Y_{13} = C$ in 3-by-3 block matrices, with all other blocks zero. Set

$$S = I + X, \quad T = I + 2X + Y.$$

We have $X^3 = 0$ and $XY = YX = Y^2 = 0$. Thus S, T are invertible, commute, and have unit diagonal, while $S^2 - T = X^2 - Y$ has sole potentially nonzero block $AB - C$. These identities use ordered ring multiplication and remain valid in characteristic two. Apply the reduction convention and Theorem 1.1. Zero padding for the first task and identity padding for the others cover every sufficiently large N . The general verifier supplies the upper bounds, using the same oracle twice in the squaring case. $\square$

A.2 Positive-definite inputs and conditioning

The next construction gives well-conditioned integer inputs and rational inputs in any prescribed neighborhood of the identity.

Corollary A.2 (Positive definiteness, conditioning, and near-identity inputs). *Let $0 < \delta_N < 1$ be any prescribed sequence. Exact verification on rational symmetric positive-definite N -by- N triples (U, V, W) has bounded-error query complexity $\Theta(N^{5/3})$ and a finite-support adversary lower bound $\Omega(N^{5/3})$, even under the promises*

$$\text{diag } U = \text{diag } V = \text{diag } W = \mathbf{1}, \quad \text{spec}(U) \cup \text{spec}(V) \cup \text{spec}(W) \subseteq [1 - \delta_N, 1 + \delta_N].$$

Every off-diagonal entry then has magnitude at most δ_N , and every spectral condition number is at most $(1 + \delta_N)/(1 - \delta_N)$. The hard entries have numerator and denominator bit lengths $O(N + \log_2(1/\delta_N))$.

The same query and adversary bounds hold for integer symmetric positive-definite triples with all three condition numbers at most $(1 + \delta_N)/(1 - \delta_N)$. Their hard entries use $O(N + \log_2(1/\delta_N))$ bits, and this integer restriction can retain a unit-singleton residual. The lower-bound constants in both families are independent of the sequence.

In particular, taking $\delta_N = 1/4$ gives the integer restriction with condition numbers at most $5/3$, and the rational restriction with unit diagonal, entries in $[-1, 1]$, and all eigenvalues in $[3/4, 5/4]$. Both fixed-parameter restrictions use $O(N)$ bits per entry. Rational normalization rescales the residual.

Proof. Fix sufficiently large N , put $n = \lfloor N/6 \rfloor$, and use the finite integer source domain $\mathcal{X}_{n,H}$ with $H = H_n$. Use the $2n$ -by- $2n$ matrices M, N_0 from the zero-product reduction in the proof of Corollary A.1. Form symmetric 3-by-3 block matrices with blocks of size $2n$:

$$P_{12} = M, \quad P_{21} = M^\top, \quad Q_{23} = N_0, \quad Q_{32} = N_0^\top,$$

with all other blocks zero. Thus P, Q are $6n$ -by- $6n$ matrices with zero diagonals and disjoint supports, and the only potentially nonzero block of PQ is

$$(PQ)_{13} = MN_0 = \text{diag}(AB - C, 0).$$

On the entire source domain, Euclidean operator norms satisfy

$$\|P\| = \|M\| \leq 4n^2H^2 + 2nH + n, \quad \|Q\| = \|N_0\| \leq 2nH + 1.$$

Put

$$K_0 = 4n^2H^2 + 4nH + n + 1, \quad L = \lceil K_0/\delta_N \rceil.$$

The integer L is input independent, and $\|P\| + \|Q\| \leq K_0 \leq \delta_N L$.

Define the unnormalized integer triple by

$$\tilde{U} = LI + P, \quad \tilde{V} = LI + Q, \quad \tilde{W} = L^2I + L(P + Q).$$

Its residual is

$$\tilde{U}\tilde{V} - \tilde{W} = PQ.$$

The eigenvalues of $\tilde{U}, \tilde{V}$ lie in $[(1 - \delta_N)L, (1 + \delta_N)L]$, and those of $\tilde{W}$ lie in $[(1 - \delta_N)L^2, (1 + \delta_N)L^2]$. Consequently all three are positive definite and have the asserted condition-number bound, on YES

and NO inputs alike. A source residual E_{ij} becomes one unit entry in the $(1, 3)$ block of PQ , so the integer construction preserves the unit-singleton promise.

Normalize this same triple by setting

$$U = \tilde{U}/L = I + P/L, \quad V = \tilde{V}/L = I + Q/L, \quad W = \tilde{W}/L^2 = I + (P + Q)/L.$$

Then

$$UV - W = PQ/L^2, \quad \max\{\|U - I\|, \|V - I\|, \|W - I\|\} \leq K_0/L \leq \delta_N.$$

These symmetric unit-diagonal matrices have the asserted spectra and are positive definite because $\delta_N < 1$. Each off-diagonal entry of $T \in \{U, V, W\}$ has magnitude at most $\|T - I\| \leq \delta_N$, so all entries lie in $[-1, 1]$. For $\delta_N = 1/4$, $L = 4K_0$.

Pad the rational triple by (I, I, I) of size $N - 6n$, or the integer triple by (LI, LI, L^2I) of that size. These direct sums preserve all the respective promises, the residual (up to zero padding), and the equivalence of target equality with $AB = C$.

Disjoint supports make every coordinate constant or injective affine in one source entry, and the complete encoding is injective. The reduction convention of Appendix A.1 and Theorems 1.3 and 1.4 give both lower bounds, including on the unit-singleton source promise, with absolute losses independent of L, δ_N . The general verifier gives the upper bounds.

At $H = H_n = 2^{100n}$, $\log_2 K_0 = O(n)$ and $L \leq 2K_0/\delta_N$, so $\log_2 L = O(N + \log_2(1/\delta_N))$. Rational source-dependent numerators have $O(n)$ bits and denominators L ; reduction to lowest terms cannot increase either. Integer entries have magnitude at most $L^2 + LK_0 \leq 2L^2$ and obey the same bit bound. Only the integer L occurs in the encoding, so δ_N need not be rational. $\square$

Both $\delta_N = 2^{-cN}$ for fixed $c > 0$ and $\delta_N = 1/N$ for $N \geq 2$ permit $O(N)$ -bit hard entries; the latter gives condition numbers at most $1 + 2/(N - 1)$, tending to one. For arbitrary smaller δ_N , the precision-dependent bit bound applies: $O(N)$ is not uniform over sequences. The integer triple has residual PQ and the same condition-number bound but need not be close to the identity. Normalization can make the residual PQ/L^2 arbitrarily small, so these are not fixed-tolerance numerical-verification lower bounds.

The positive-definite and commuting restrictions are separate: the positive-definite construction does not promise commutation on NO inputs. Positive definiteness also does not mean entrywise positivity.

A.3 Strictly positive entries

Corollary A.3 (Strictly positive entries). *Exact verification on strictly positive integer matrices, and on rational matrices with every entry in $(0, 1]$, has bounded-error query complexity $\Theta(N^{5/3})$ and adversary lower bound $\Omega(N^{5/3})$. Both conclusions have hard restrictions using $O(N)$ bits per entry.*

Proof. Let J be the all-ones n -by- n matrix and $m = 2H + 1$. Define

$$A_+ = \begin{pmatrix} mJ + A & mJ - A \\ mJ + A & mJ - A \end{pmatrix}, \quad B_+ = \begin{pmatrix} mJ + B & mJ + B \\ mJ - B & mJ - B \end{pmatrix},$$

and let every block of C_+ equal $2C + 2nm^2J$. Direct multiplication gives

$$A_+B_+ - C_+ = 2 \begin{pmatrix} AB - C & AB - C \\ AB - C & AB - C \end{pmatrix}.$$

The factor entries are at least one; the entries of C_+ are at least $8nH + 2n - 2 > 0$. All entries are at most $R_0 = 4nm^2 + 2$, and all coordinate maps are injective affine maps over the integers. This proves the even-dimensional lower bound.

For odd dimension, put $d = 2n$ and $a = dm$. The identities $A_+ \mathbf{1} = a \mathbf{1}$ and $\mathbf{1}^\top B_+ = a \mathbf{1}^\top$ allow the strictly positive padding

$$\hat{A} = \begin{pmatrix} A_+ & \mathbf{1} \\ \mathbf{1}^\top & 1 \end{pmatrix}, \quad \hat{B} = \begin{pmatrix} B_+ & \mathbf{1} \\ \mathbf{1}^\top & 1 \end{pmatrix}, \quad \hat{C} = \begin{pmatrix} C_+ + J_d & (a+1)\mathbf{1} \\ (a+1)\mathbf{1}^\top & d+1 \end{pmatrix},$$

where J_d is the d -by- d all-ones matrix and the vectors have length d . The residual is $\text{diag}(A_+ B_+ - C_+, 0)$; all entries are at most $R_0 + 1$. No unknown row or column sum is queried.

For either parity, write the resulting triple as (A', B', C') and let $R = R_0$ or $R_0 + 1$, respectively. The triple $(A'/R, B'/R, C'/R^2)$ has entries in $(0, 1]$ and preserves equality. All encodings still require $O(n)$ bits when $H = H_n$. Theorems 1.3 and 1.4, together with the general upper bound, prove the claims. A source unit error becomes four nonzero residual entries of value 2 before normalization, and $2/R^2$ afterward. Thus this reduction does not assert a positive-entry unit-singleton theorem. $\square$

A.4 Quadratic identities and comparison of products

Matrix-set commutativity under entry access was studied by Itakura [13]; here we consider two matrices with additional structural promises, together with other quadratic identities.

Corollary A.4 (Quadratic verification tasks). *Over every fixed nonzero unital associative ring, each of the following tasks on N -by- N matrices has bounded-error exact whole-entry query complexity $\Theta(N^{5/3})$ and a finite-support negative-weight adversary lower bound $\Omega(N^{5/3})$:*

1. *testing idempotence, $H^2 = H$, even when H is upper triangular and its diagonal entries are prescribed zeros and ones;*
2. *testing $Z^2 = 0$, even when Z is strictly upper triangular and is promised to satisfy $Z^3 = 0$;*
3. *testing whether two matrices commute, even under either of the separate promises that both are idempotent, or that both are upper triangular with unit diagonal and satisfy $(U - I)^2 = (V - I)^2 = 0$;*
4. *comparing two products, $A_1 B_1 = A_2 B_2$.*

The implicit constants may depend on the ring; over finite fields they are uniform in the field size.

Proof. Let (A, B, C) be an arbitrary n -by- n source instance and put $D = AB - C$. For idempotence, use

$$H = \begin{pmatrix} 0 & A & C \\ 0 & I & B \\ 0 & 0 & 0 \end{pmatrix}, \quad H^2 - H = \begin{pmatrix} 0 & 0 & D \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus $H^2 = H$ if and only if $AB = C$.

For the next two tasks, define 4-by-4 block matrices X, Y by

$$X_{12} = A, \quad X_{13} = -C, \quad Y_{24} = B, \quad Y_{34} = I,$$

with all other blocks zero. Ordered block multiplication gives

$$X^2 = Y^2 = YX = 0, \quad (XY)_{14} = D,$$

and every other block of XY is zero. Hence $Z = X + Y$ is strictly upper triangular, $Z^2 = XY$, and $Z^3 = 0$. Testing $Z^2 = 0$ therefore recovers the source problem.

To obtain two idempotents, put

$$E = \text{diag}(I, 0, 0, 0), \quad F = \text{diag}(0, 0, 0, I), \quad P = E + X, \quad Q = F + Y.$$

The block positions give

$$EX = X, \quad XE = 0, \quad YF = Y, \quad FY = 0, \quad EF = FE = EY = YE = XF = FX = 0.$$

Consequently $P^2 = P$, $Q^2 = Q$, $QP = 0$, and $PQ = XY$. Thus these two promised idempotents commute exactly when $D = 0$. For the separate unit-diagonal restriction, take $U = I + X$, $V = I + Y$. Their inverses are $I - X$, $I - Y$, and

$$(U - I)^2 = (V - I)^2 = 0, \quad UV - VU = XY.$$

All these identities remain valid in characteristic two.

Comparison of two products contains the source problem as the restriction $(A_1, B_1, A_2, B_2) = (A, B, C, I)$. Every encoding is injective and uses only constants, source entries, or their negatives. The reduction convention of Appendix A.1 therefore transfers the finite adversary witnesses of Theorem 1.1 without loss. Dimensions grow by at most a factor of four. Zero padding handles the first two tasks and idempotent commutativity; identity padding handles unit-diagonal commutativity. Theorems 1.2 and 1.4 give field-uniform constants.

For the upper bounds, use the matrix-product verifier on (H, H, H) or $(Z, Z, 0)$. Comparison of two products reduces to zero-product verification through

$$\begin{pmatrix} A_1 & -A_2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} B_1 & 0 \\ B_2 & 0 \end{pmatrix} = \begin{pmatrix} A_1 B_1 - A_2 B_2 & 0 \\ 0 & 0 \end{pmatrix}.$$

Each query to these $2N$ -by- $2N$ factors costs $O(1)$ input queries, so Theorem 1.1 gives $O(N^{5/3})$ queries. Commutativity is the special case $(A_1, B_1, A_2, B_2) = (P, Q, Q, P)$. $\square$

Here idempotence means the algebraic identity $H^2 = H$, with no self-adjointness requirement. Likewise, the square-zero reduction tests whether the nilpotency index is at most two: all constructed instances are already nilpotent because $Z^3 = 0$.

Remark A.5 (Simultaneous diagonalizability of idempotents). Over a fixed field K , two idempotents P, Q are simultaneously diagonalizable exactly when they commute. Indeed, in the commuting case the four mutually annihilating idempotents

$$PQ, \quad P(I - Q), \quad (I - P)Q, \quad (I - P)(I - Q)$$

sum to I . Their images form a direct sum on each of which both P, Q act as scalars in $\{0, 1\}$, giving a common eigenbasis. The converse is immediate. Corollary A.4 therefore gives query complexity $\Theta(N^{5/3})$ and a finite-support adversary lower bound $\Omega(N^{5/3})$ for testing this property on two promised idempotents.

This last task tests simultaneous diagonalizability; it does not settle product verification when all supplied matrices are promised to be simultaneously diagonalizable.

A.5 Gram comparisons and finite-field orthogonality

For matrices over a field, $*$ will mean ordinary transpose, except when complex conjugate transpose is explicitly specified. These two operations lead to different conclusions for single-Gram verification.

Corollary A.6 (Comparing Gram matrices). *Over every fixed field, the problem*

$$F^\top F = G^\top G$$

for two N -by- N input matrices has bounded-error exact entry-query complexity $\Theta(N^{5/3})$ and a finite-support negative-weight adversary lower bound $\Omega(N^{5/3})$. The same conclusions hold over $\mathbb{C}$ with conjugate transpose in place of transpose. The finite-field constants are uniform in the field size.

Proof. Given a source triple (A, B, C) of dimension n , form

$$F_0 = \begin{pmatrix} A^* & B \\ I & -C \end{pmatrix}, \quad G_0 = \begin{pmatrix} A^* & 0 \\ 0 & B \\ I & 0 \\ 0 & C \end{pmatrix}.$$

The matrices have $2n$ columns, and direct multiplication gives

$$F_0^* F_0 - G_0^* G_0 = \begin{pmatrix} 0 & AB - C \\ (AB - C)^* & 0 \end{pmatrix}.$$

Indeed, the diagonal blocks of both Gram matrices are $AA^* + I$ and $B^*B + C^*C$, respectively. Thus equality of the two Gram matrices is equivalent to $AB = C$.

Zero-pad F_0 by rows and both matrices by columns to dimension $4n$, and further to any sufficiently large target dimension. Each coordinate is constant or injective in one source entry, and F_0 determines A, B, C . The reduction convention of Appendix A.1 and Theorem 1.1 give both lower bounds; Theorems 1.2 and 1.4 give finite-field uniformity.

For the upper bound, Corollary A.4 compares F^*F and G^*G with constant query overhead for transposed or conjugate-transposed entries. $\square$

The following constant-dimensional construction uses ordinary transpose and works over every finite field.

Lemma A.7 (An isotropic pair over every finite field). *For every finite field K , there is an invertible matrix $T = [u \ v \ w] \in K^{3 \times 3}$ and a nonzero $\alpha \in K$ such that*

$$T^\top T = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & \alpha \end{pmatrix}.$$

The matrix T depends only on the field, not on the input.

Proof. In characteristic two, take

$$u = (1, 1, 0)^\top, \quad v = (0, 1, 1)^\top, \quad w = (1, 1, 1)^\top.$$

Their Gram matrix has the asserted form with $\alpha = 1$.

Suppose the characteristic is odd and $|K| = q$. The set of squares has $(q+1)/2$ elements, so it intersects minus one minus that set. Consequently there are $a, b \in K$ with $a^2 + b^2 = -1$. Put

$$u = (a, b, 1)^\top, \quad v = (0, 0, 1)^\top - u/2, \quad w = (b, -a, 0)^\top.$$

Then

$$u^\top u = v^\top v = 0, \quad u^\top v = 1, \quad u^\top w = v^\top w = 0, \quad w^\top w = -1.$$

This gives $\alpha = -1$. In both cases the displayed Gram matrix is nonsingular, which also proves that T is invertible. $\square$

Corollary A.8 (Transpose-Gram zero and orthogonality). *Over every finite field, and over $\mathbb{C}$ with ordinary transpose, each of the following problems has bounded-error exact entry-query complexity $\Theta(N^{5/3})$ and a finite-support negative-weight adversary lower bound $\Omega(N^{5/3})$:*

$$F^\top F = 0, \quad G^\top G = I,$$

where the input matrix is N -by- N and the target is known. For the second problem, the lower bound holds even when G is promised to be invertible and to satisfy $(G - I)^3 = 0$. For finite fields, all constants are uniform in the field size.

Proof. First let the field be finite and fix u, v, w, T, α from Lemma A.7. For the first problem, use the d -by- d matrices M, N_0 from the proof of Corollary A.1, where $d = 2n$. They satisfy $MN_0 = \text{diag}(AB - C, 0)$. Define the $3d$ -by- $2d$ matrix $F_0 = [u \otimes M^\top \quad v \otimes N_0]$. The identities for u, v give

$$F_0^\top F_0 = \begin{pmatrix} 0 & MN_0 \\ (MN_0)^\top & 0 \end{pmatrix}.$$

Thus its Gram matrix is zero exactly when $AB = C$. Add d zero columns to make the matrix square of dimension $6n$. Every target entry is a known scalar multiple of one source entry or a constant. Since u, v are nonzero, F_0 determines M, N_0 , and hence the source triple.

For the second problem, use the upper-triangular inverse-test matrices U, V from the same proof. They have dimension $d = 3n$, unit diagonal, and satisfy

$$VU = I \iff AB = C.$$

Set

$$S = T \otimes I_d, \quad P = \text{diag}(U, V^\top, I_d), \quad G = SPS^{-1}.$$

Writing

$$H = S^\top S = \begin{pmatrix} 0 & I_d & 0 \\ I_d & 0 & 0 \\ 0 & 0 & \alpha I_d \end{pmatrix},$$

we obtain

$$P^\top H P = \begin{pmatrix} 0 & (VU)^\top & 0 \\ VU & 0 & 0 \\ 0 & 0 & \alpha I_d \end{pmatrix}.$$

Therefore

$$G^\top G = I \iff P^\top H P = H \iff VU = I \iff AB = C.$$

Both U, V are always invertible, and $(U - I)^3 = (V^\top - I)^3 = 0$. Consequently G is always invertible and $(G - I)^3 = 0$.

The conjugation remains entry-local. More explicitly,

$$G_{(a,i),(b,j)} = T_{a1}(T^{-1})_{1b}U_{ij} + T_{a2}(T^{-1})_{2b}(V^\top)_{ij} + T_{a3}(T^{-1})_{3b}\delta_{ij}.$$

For $i < j$, only the first matrix entry can be nonconstant; for $i > j$, only the second can be nonconstant; and for $i = j$ all three are constants. This uses the ordinary upper-triangular forms of U, V , including their identity diagonal blocks. Hence each target entry depends on at most one source entry. The encoding is injective because conjugation is invertible and U contains A, B, C .

In both reductions, every nonconstant coordinate map is injective. The reduction convention of Appendix A.1 and Theorems 1.2 and 1.4 therefore give the lower bounds with uniform constants. Zero padding extends the first construction, and identity padding extends the second, to every sufficiently large N ; the latter also preserves $(G - I)^3 = 0$. The general verifier applies to $(F^\top, F, 0)$ and $(G^\top, G, I)$ with constant query overhead, proving the matching upper bounds.

Over $\mathbb{C}$, take $a = i$, $b = 0$ in the odd-characteristic formulas of Lemma A.7, where $i^2 = -1$. Their identities give an invertible T with the same Gram matrix and $\alpha = -1$. The same constructions, padding, and upper bounds apply; Theorem 1.1 and the reduction convention supply the finite-support lower bound. $\square$

Transpose versus adjoint. These single-Gram conclusions do not apply to the positive-definite real or complex Hermitian inner product. For real transpose or complex conjugate transpose,

$$F^*F = 0 \iff F = 0,$$

because every diagonal Gram entry is a sum of squared absolute values. Thus this fixed-zero problem has query complexity $\Theta(N)$: quantum search over the N^2 entries gives the upper bound, and the restriction to entries in $\{0, 1\}$ gives the matching search lower bound [4]. The comparison of two Gram matrices remains covered by Corollary A.6. No reduction above establishes the corresponding lower bound for real orthogonality, complex unitarity, or a single real/Hermitian Gram matrix with an arbitrary supplied target.

A.6 Error localization

Proof of Corollary 7.2. The lower bound follows from decision. For the upper bound, use the rectangular-verification and binary-search idea of Buhrman and Špalek [6, Section 6.1 and Appendix D]. For a t -by- t output block with inner dimension n and $t \leq 2n$, the stored-submatrix bound (47), with $k = n$, gives $O(nt^{2/3})$ queries, including the direct-reading case $t < 8$.

Pad the output to a square of power-of-two side less than $2n$, using constant zero rows and columns. Test the whole output and return **equal** if that test declares equality. Otherwise, recursively test the four quadrants and retain one declared erroneous. At depth ℓ , make each test's error at most $2^{-\ell-8}$ by $O(\ell + 1)$ repetitions. There are at most four tests per depth, so the total error probability is at most $4 \sum_{\ell \geq 0} 2^{-\ell-8} < 1/3$, also for the adaptively chosen blocks. If no quadrant is declared erroneous, return **equal**. At a leaf, read its dot product exactly: return its index if it is wrong, and **equal** otherwise. The query cost, including this final $O(n)$ check, is

$$O\left(n^{5/3} \sum_{\ell \geq 0} (\ell + 1) 2^{-2\ell/3} + n\right) = O(n^{5/3}).$$

No false witness is output; all failures to locate a genuine error are contained in the union of the verification-error events. $\square$

Use of AI: The author started with this prompt to solve the *finite fields case* of matrix-product verification: “Use polynomial method. Average degree- L acceptance polynomial over random matrix factors and restrict its residual Fourier spectrum to $O(L/n)$ active rows and columns. Use element-distinctness hardness to force it to $\Omega(n^{2/3})$ and derive $L = \Omega(n^{5/3})$ ”. ChatGPT generated an outline. The author fixed some missing gaps and iteratively verified and expanded the proofs with ChatGPT and prompted it to work out the details of adversary bounds and also apply the techniques to other input types and variants of the *matrix-product verification problem*. The author is solely responsible for the correctness of the proofs.

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